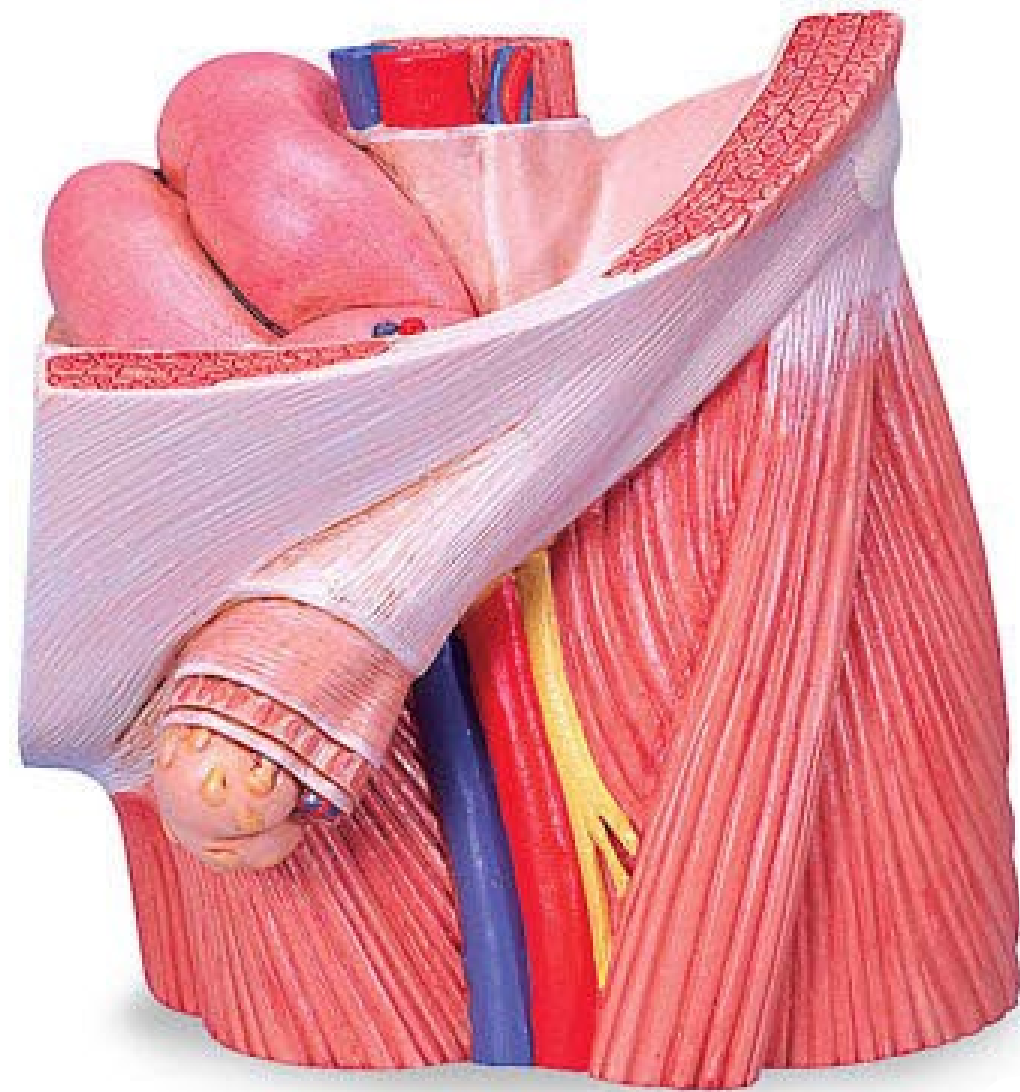




Inguinal & Scrotal Swelling

Dr. Jeremy Yip

MBBS (HK), FRCSEd (Gen)
FCSHK, FHKAM (Surgery)

Specialist in General Surgery
Honorary Clinical Assistant Professor (HKU)



50/F Admitted through AED

Left groin painful swelling with associated abdominal distension and vomiting.





30/M Bodybuilder complains of right groin pain when doing swats.



70/M complains of a bulge over the left groin



28/M has been trying to get wife pregnant unsuccessfully for the past year
and also complains of a left scrotal swelling

Groin

Painful Lumps

- Tender lymph nodes
- Strangulated inguinal hernia
- Strangulated femoral hernia
- Undescended testis
- Vascular
 - Femoral artery aneurysm
 - Saphena varix

Painless Lumps

- Skin swelling
- Non-tender lymph nodes
- Inguinal hernia
- Femoral hernia

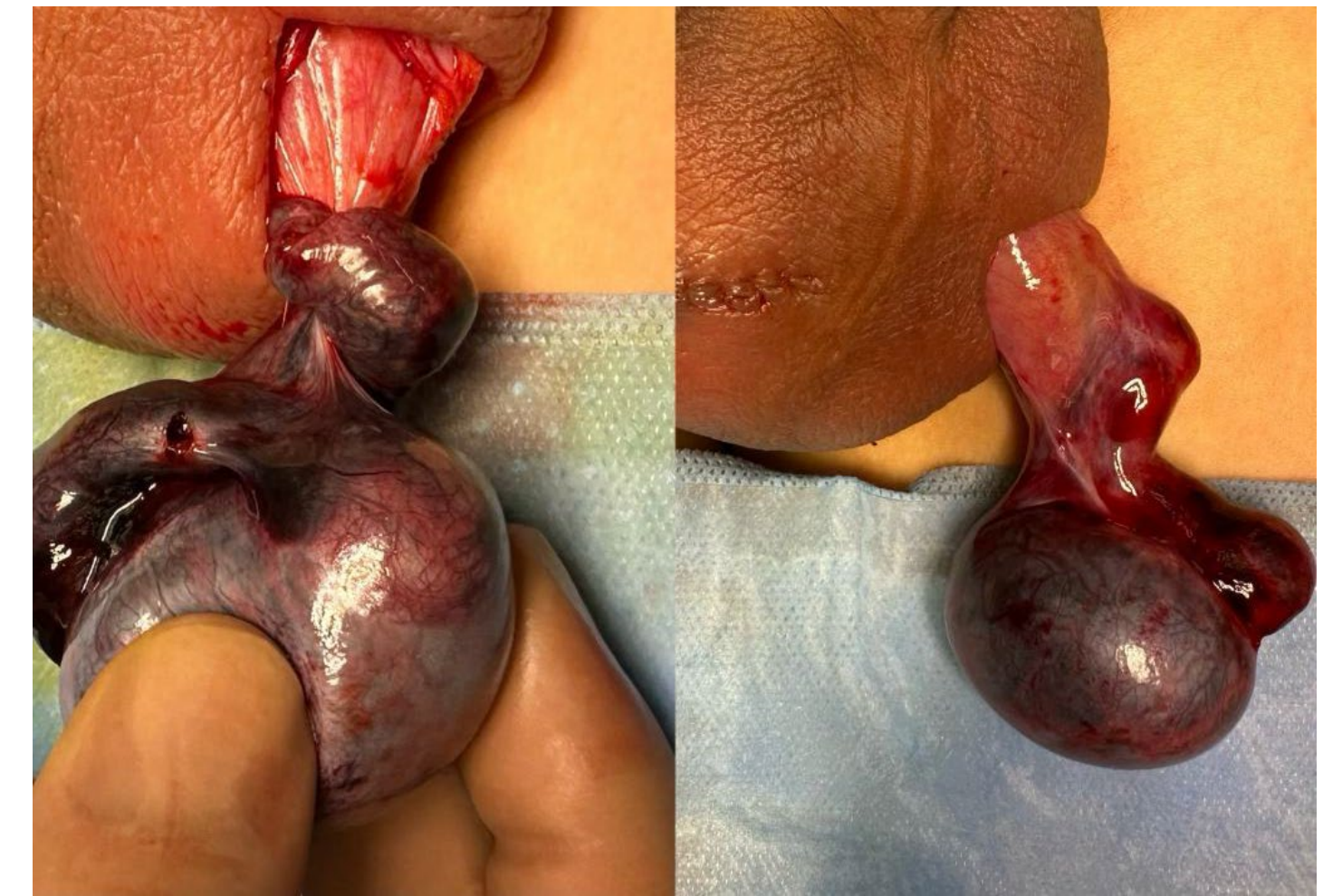
Scrotal

Painful Lumps

- Torsion of testis
- Torsion of a testicular or epididymal appendage
- Epididymo-orchitis or orchitis
- Strangulated inguinal hernia
- Hematocele / Hematoma

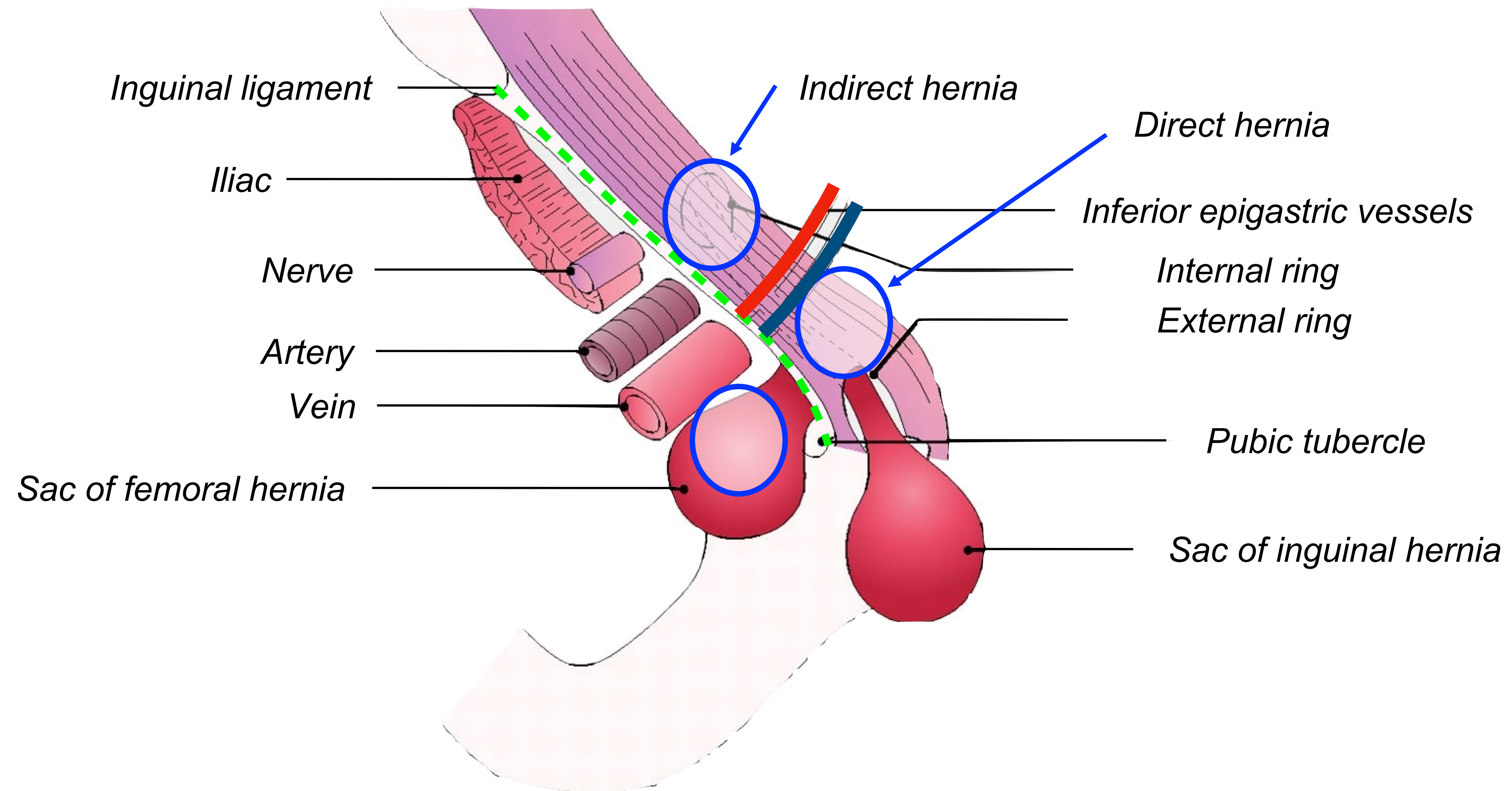
Painless Lumps

- Inguinal hernia
- Hydrocele
- Epididymal cyst
- Spermatocele
- Varicocele
- Testicular tumor
- Skin swelling

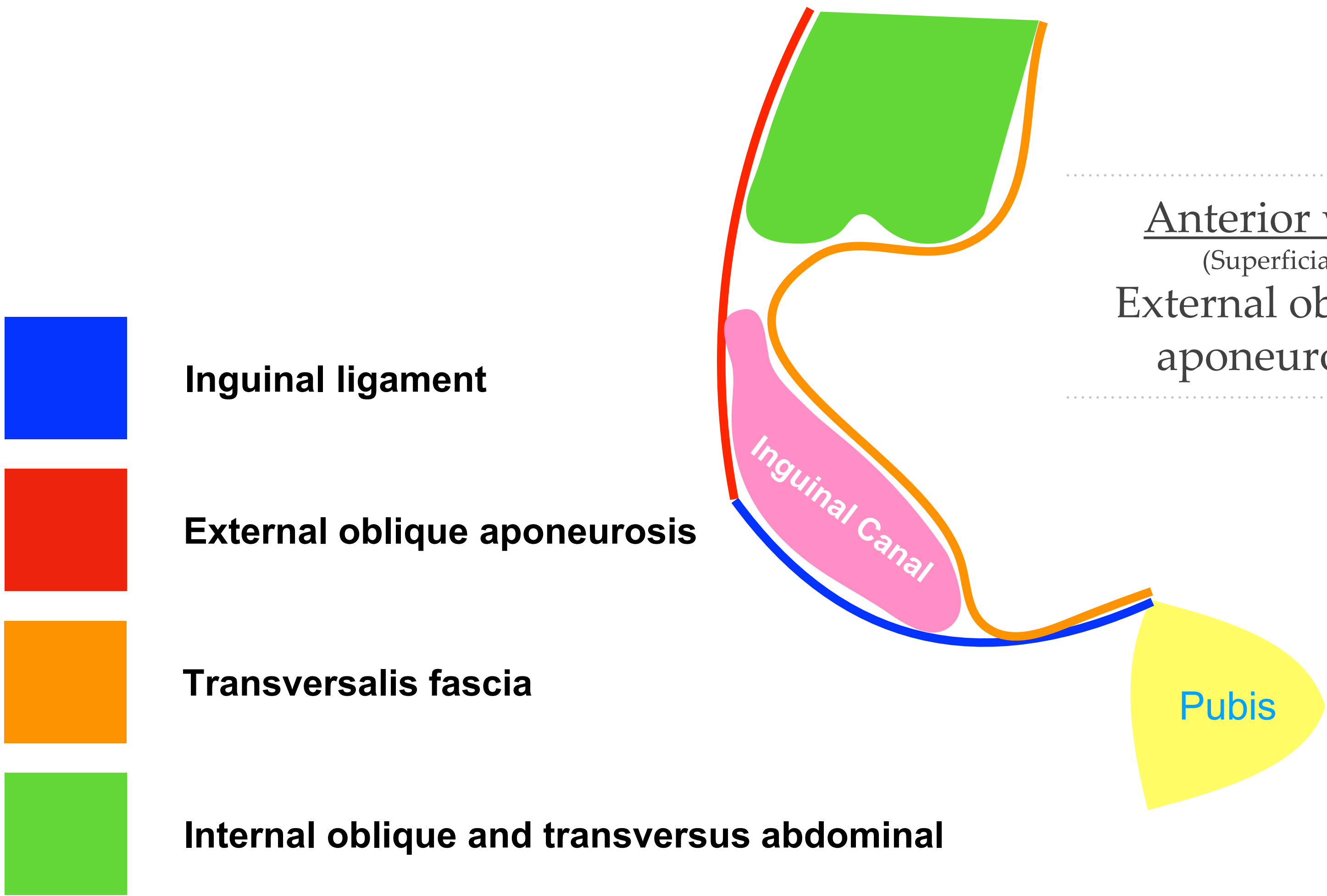


Inguinal	Inguinoscrotal	Femoral	Inguinofemoral	Scrotal
Inguinal hernia	Inguinal hernia	Femoral hernia	Inguinal lymph nodes	Skin: boils, sebaceous cyst, papillomas, warts
Lymph nodes	Hydrocele: encysted hydrocele of the cord, infantile hydrocele, hydrocele of the hernia sac	Lymph nodes	Distended psoas bursa	Subcut. tissue: lymph scrotum, filariasis [SEP] Tunica vaginalis: hydrocele, pyocele, hematocele, chylocele
Encysted hydrocele of the cord	Spermatic cord: varicocele, funiculitis, lymph vary, diffuse lipoma of the cord, hecatombs of the cord	Saphena varies	Effusion in the hip joint	Testis: Orchitis (acute/chronic), neoplasms
Undescended testis	Testis: undescended, ectopic testis	Ectopic testis		Epididymis: cysts, acute or chronic infections
In females or pregnant women: round ligament varicoses				Spermatic cord: varicocele, lymph varix

Anatomy



Inguinal Canal



Superior wall (Roof)

(Cranial)

Internal oblique
Transversus
abdominus

Anterior wall

(Superficial)

External oblique
aponeurosis

Posterior wall

Transversalis fascia
Conjoint tendon

Inferior wall (Floor)

(Caudal)

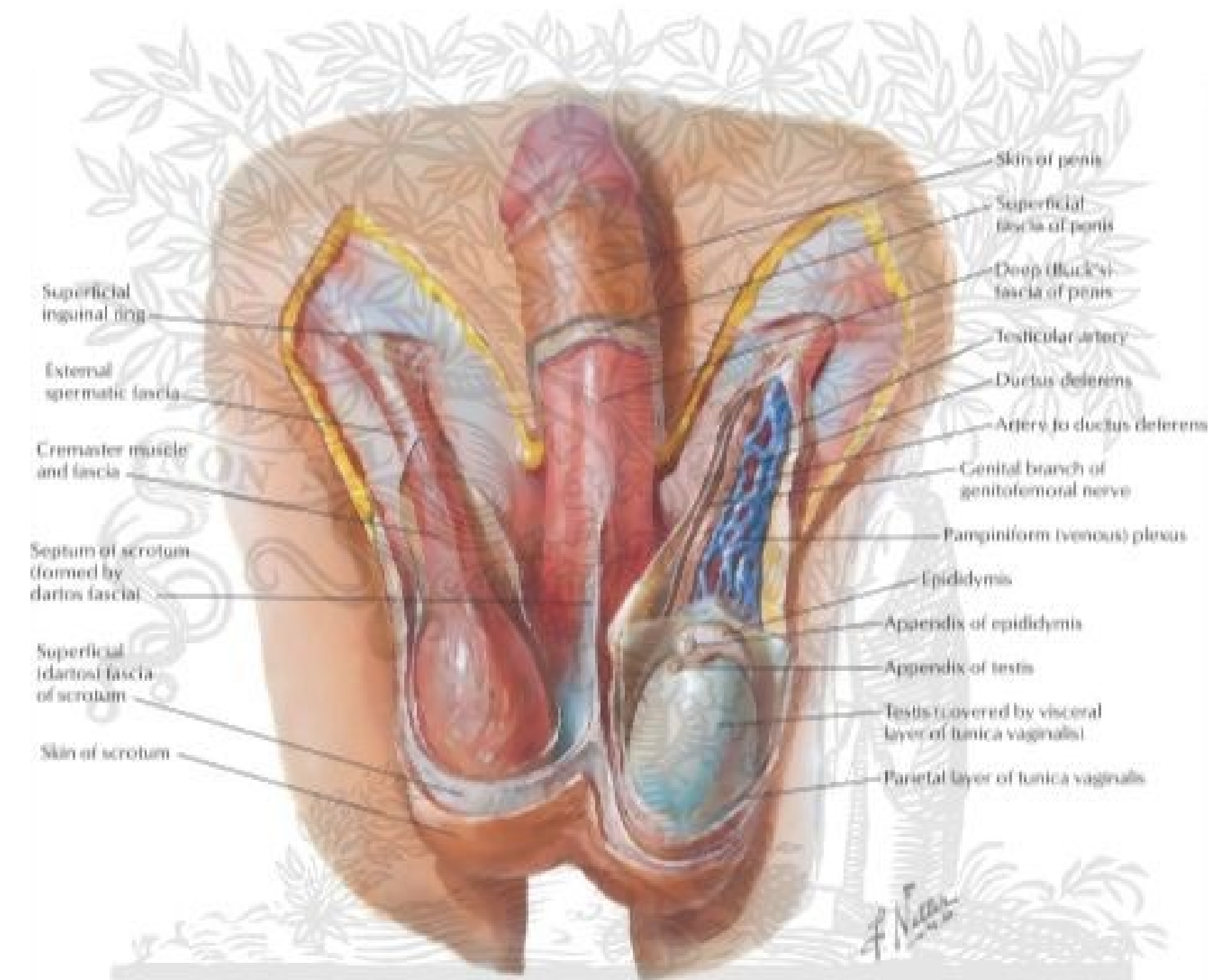
Inguinal ligament

Inguinal Canal
Male (Spermatic cord + IGN)
Female (Round ligament + IGN)

IGN: ilioinguinal nerve

Structures of the Inguinal Canal

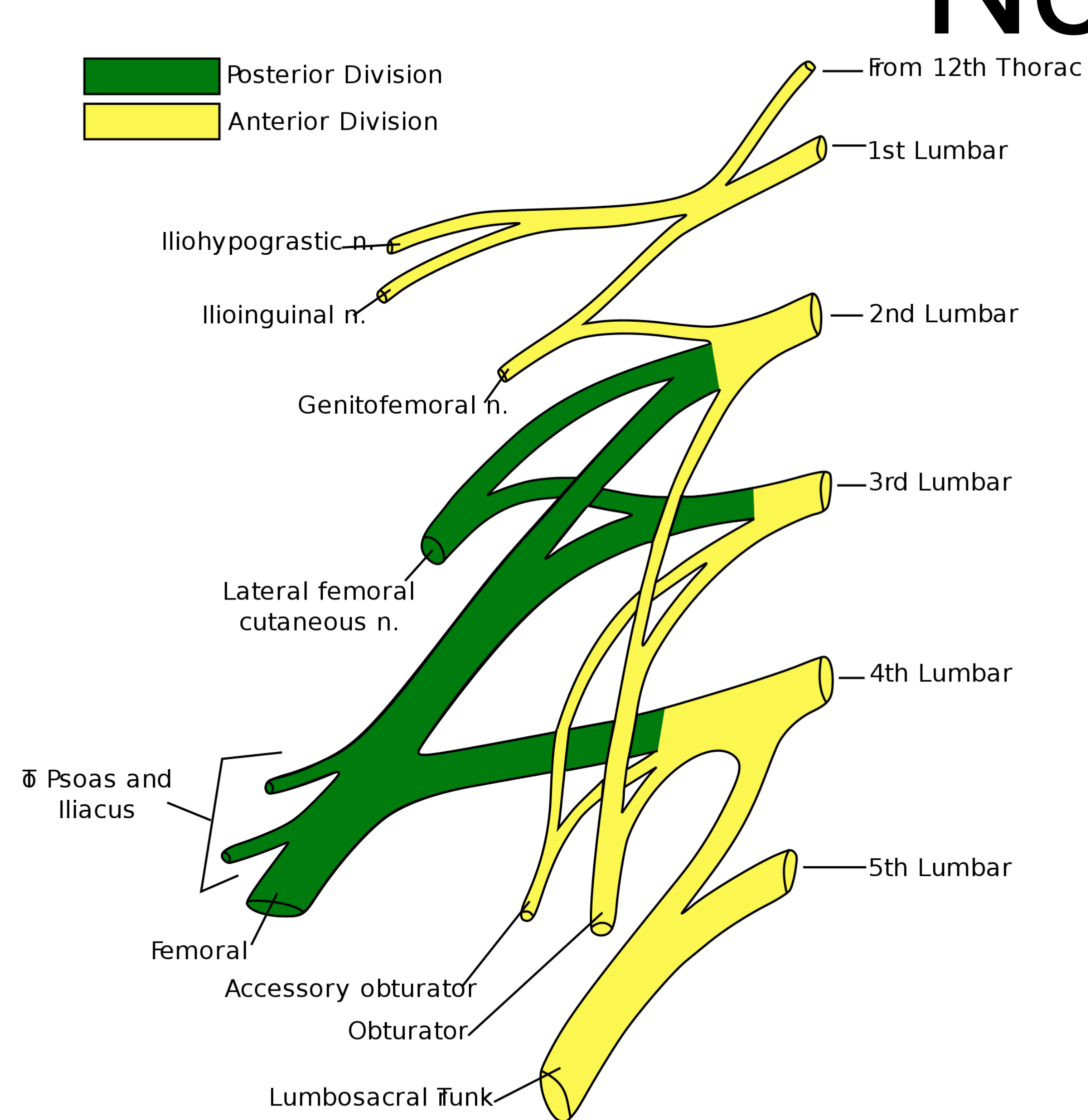
- Vas deferens (or round ligament)
- Artery to the vas
- Gonal artery
- Cremasteric artery and vein
- Gonal vein (pampiniform plexus)
- Genital branch of the genitofemoral nerve
- Ilioinguinal nerve
- Sympathetic nerves
- Lymphatics
- Internal spermatic fascia
- Cremasteric fascia and muscle
- External spermatic fascia



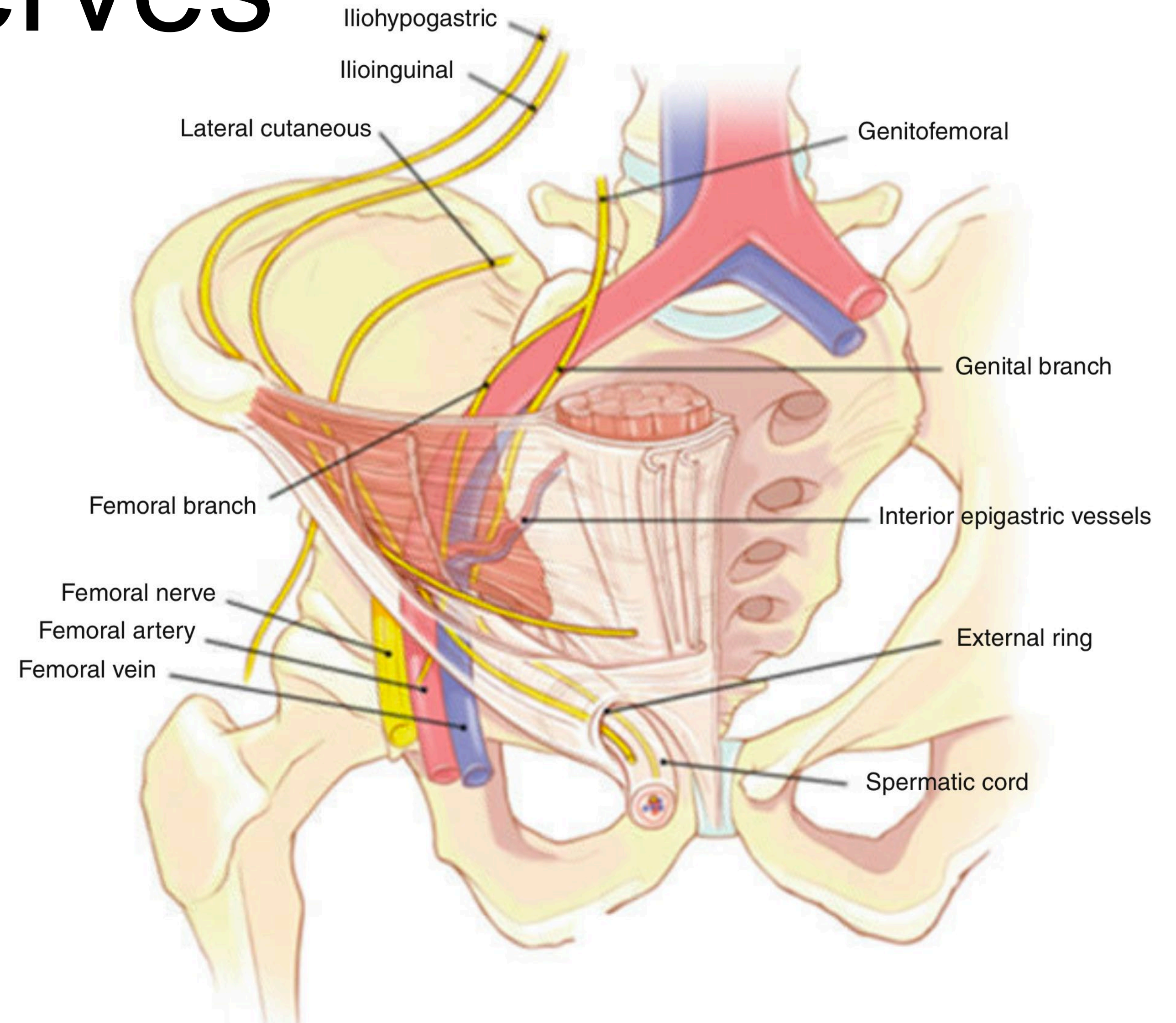
SCROTUM AND TESTES
ELSEVIER

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Nerves



https://commons.wikimedia.org/wiki/File:Lumbar_plexus.svg



Internal Ring

Male: opening in the transversalis fascia through which the vas deferens enters on its course from the pelvis.

Female: transmits the round ligament which suspends the uterus anteriorly and the genital branch of the genitofemoral nerve

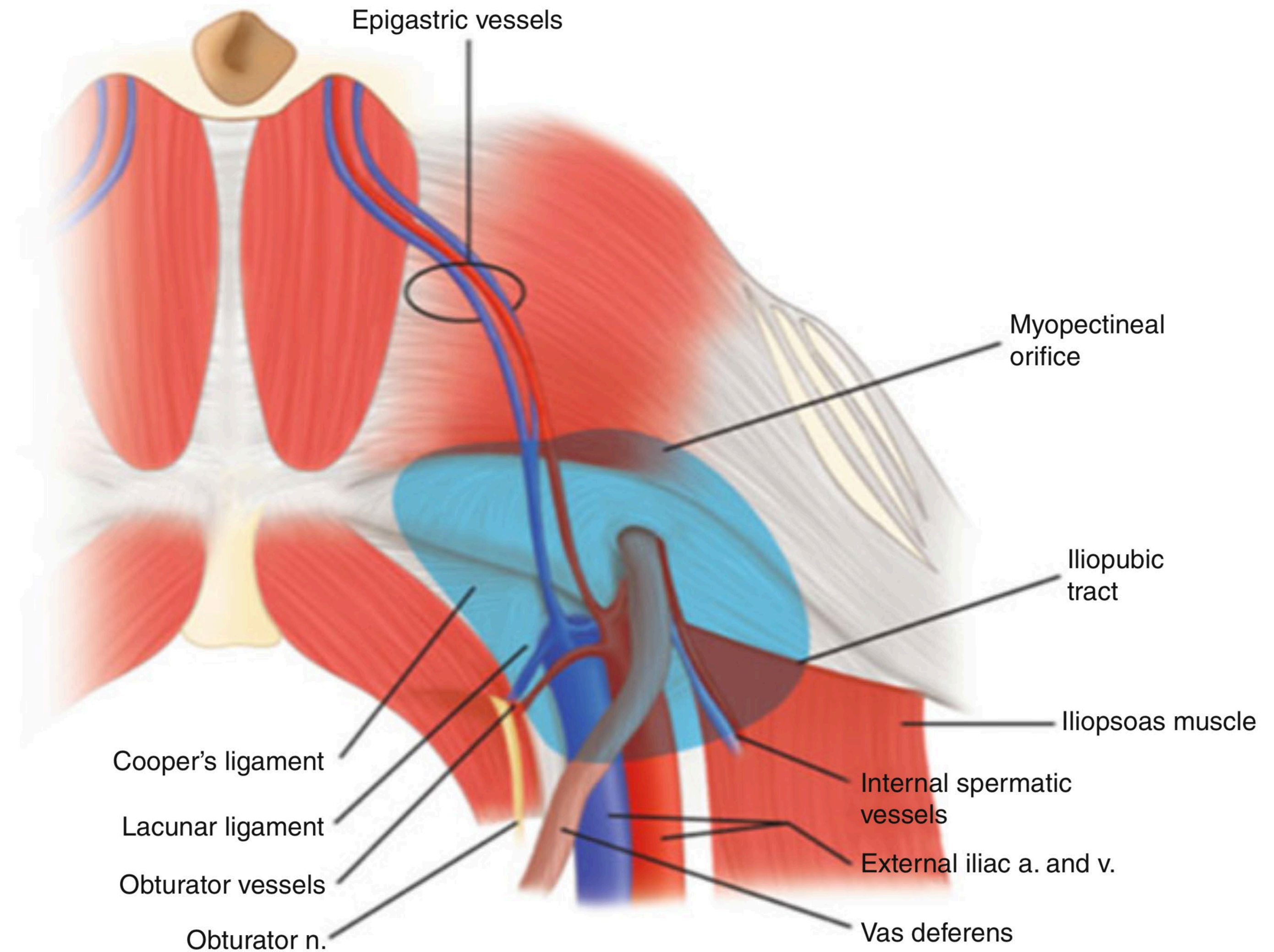
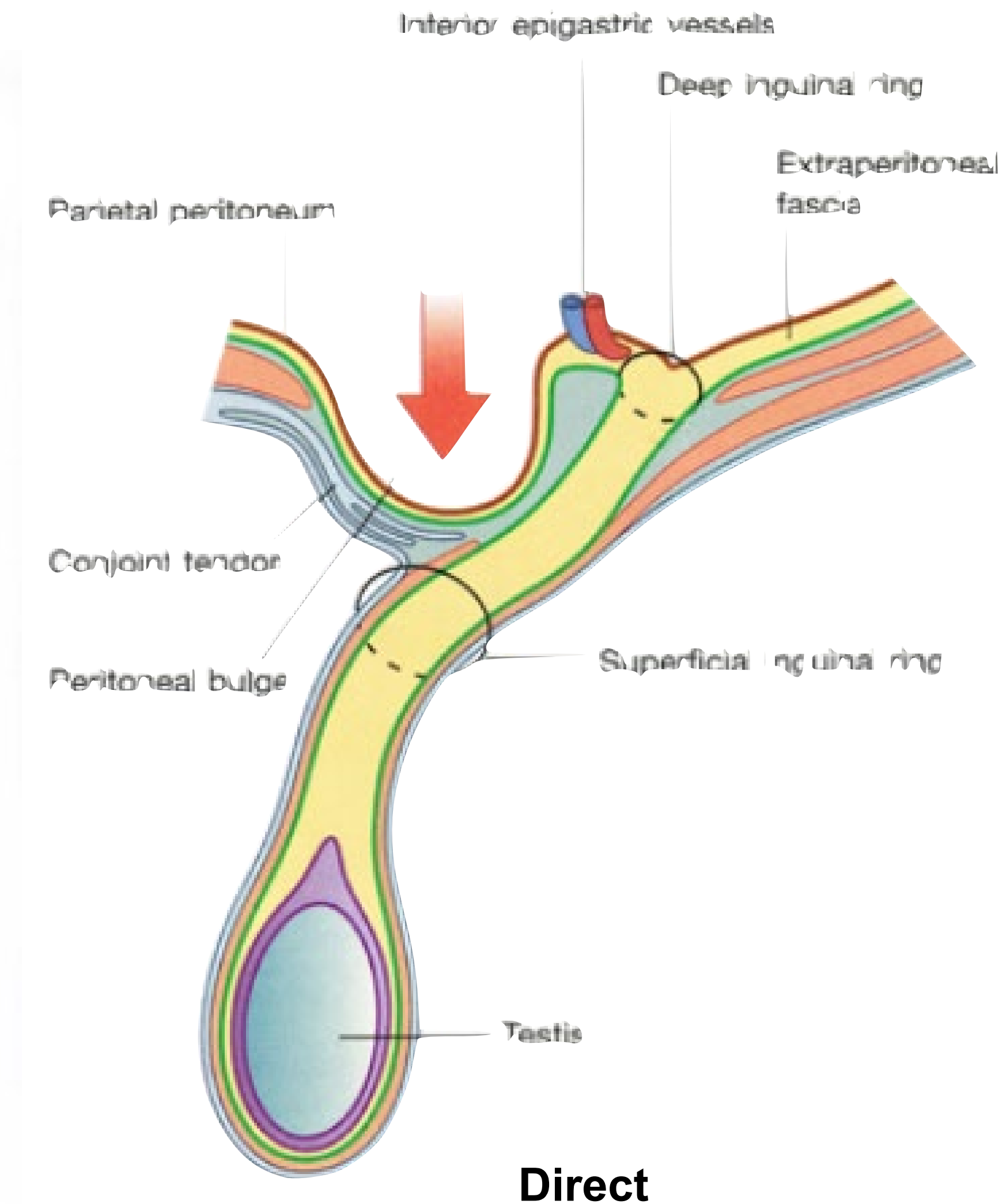
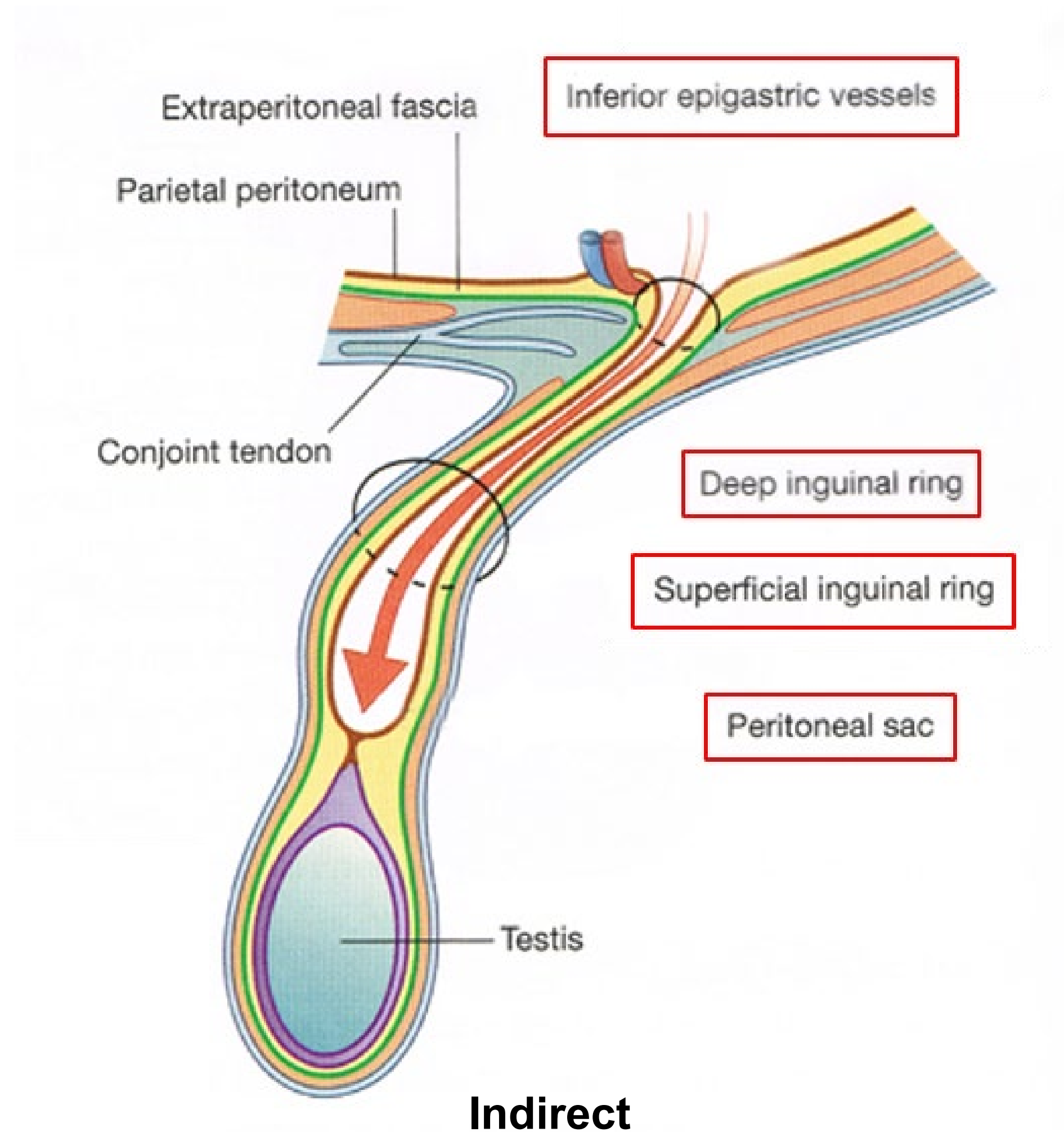


Fig. 2.2. Laparoscopic view of the internal ring (from Wagner et al. [1], with kind permission © McGraw-Hill Education).

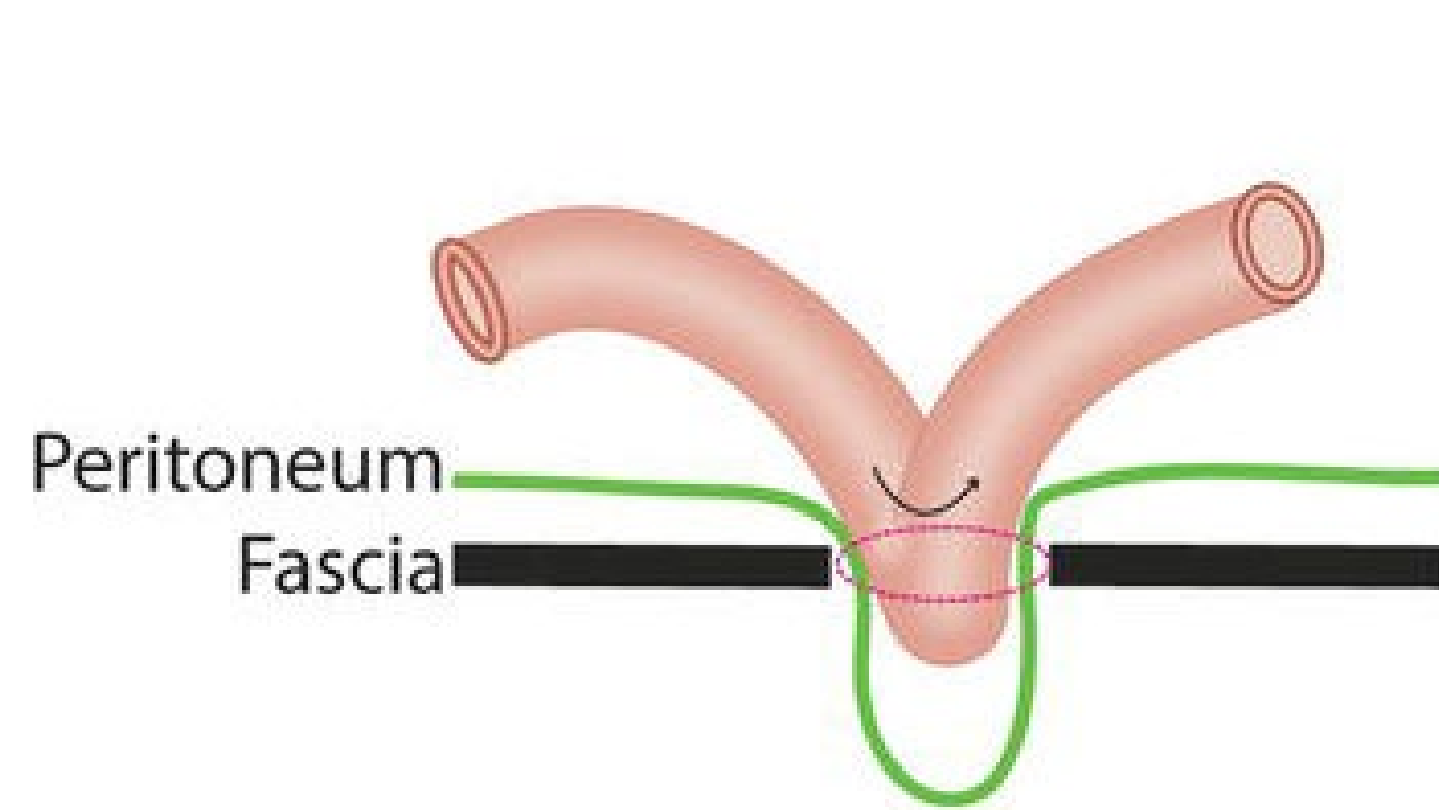
Inguinal Hernia



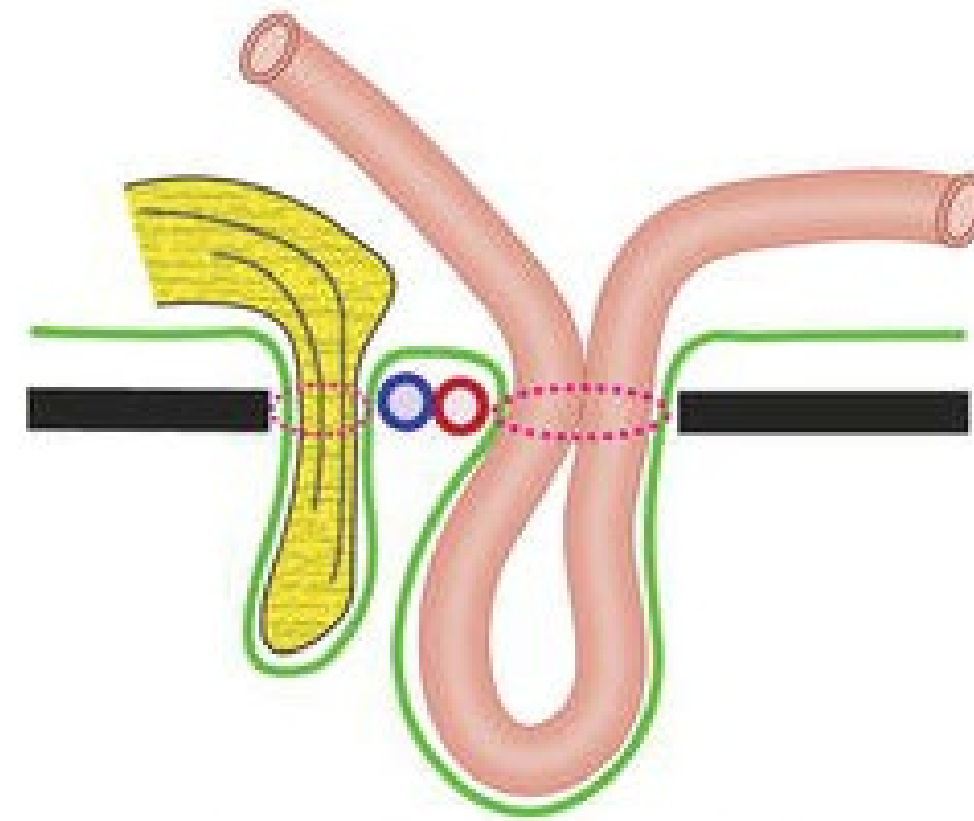
Inguinal Hernia

- **Indirect** - lateral to the inferior epigastric vessels, due to recannulation of the obliterated vaginal process.
- **Direct** - medial to the inferior epigastric vessels, due to an insufficient transversalis fascia
- **Pantaloon** - presence of both direct and indirect hernia
- **Sliding** - an internal organ forming part of the hernia wall, usually the sigmoid colon

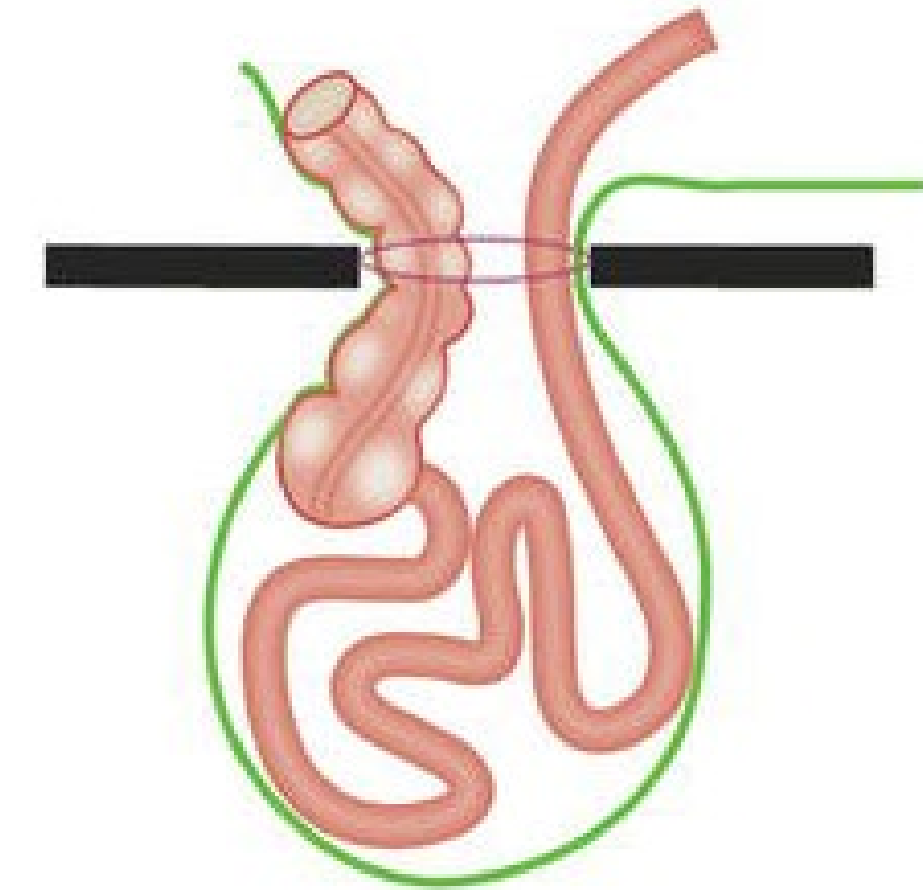
Types of Hernias



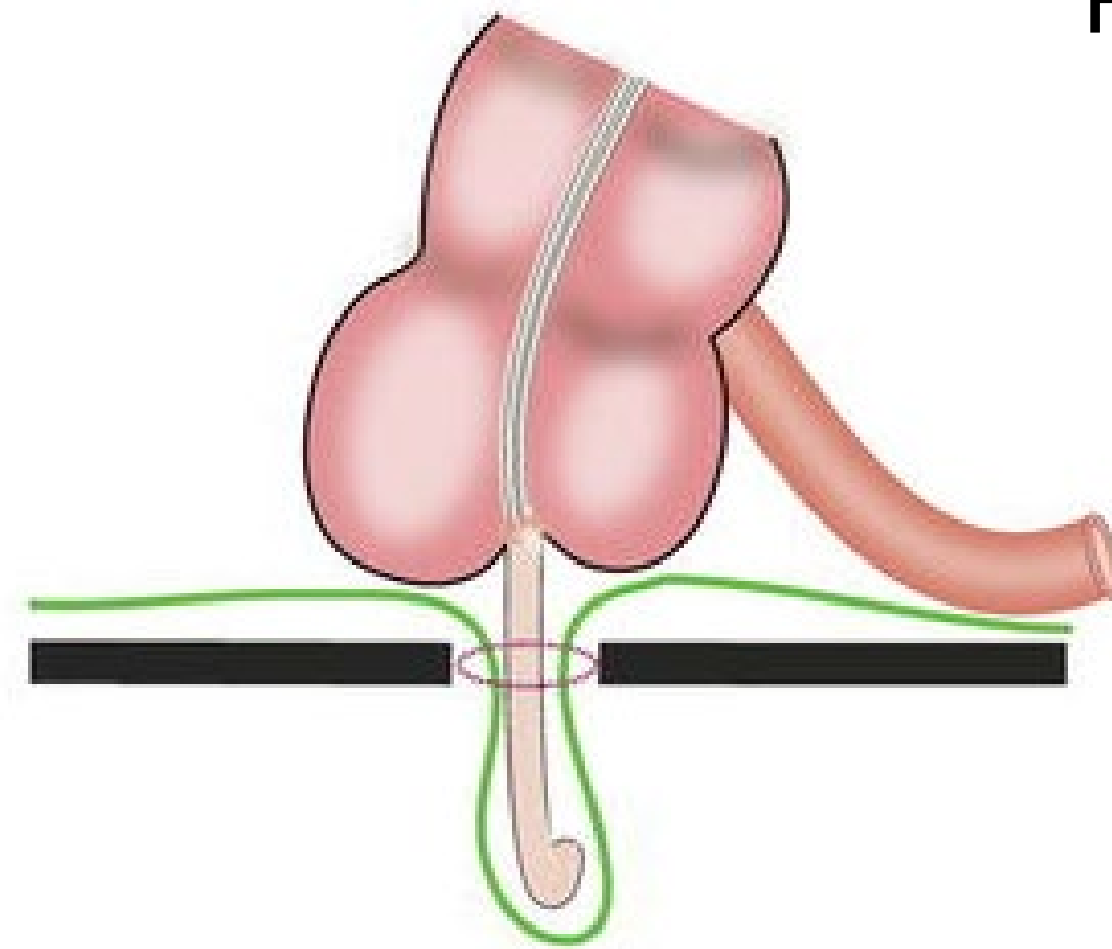
Richter's



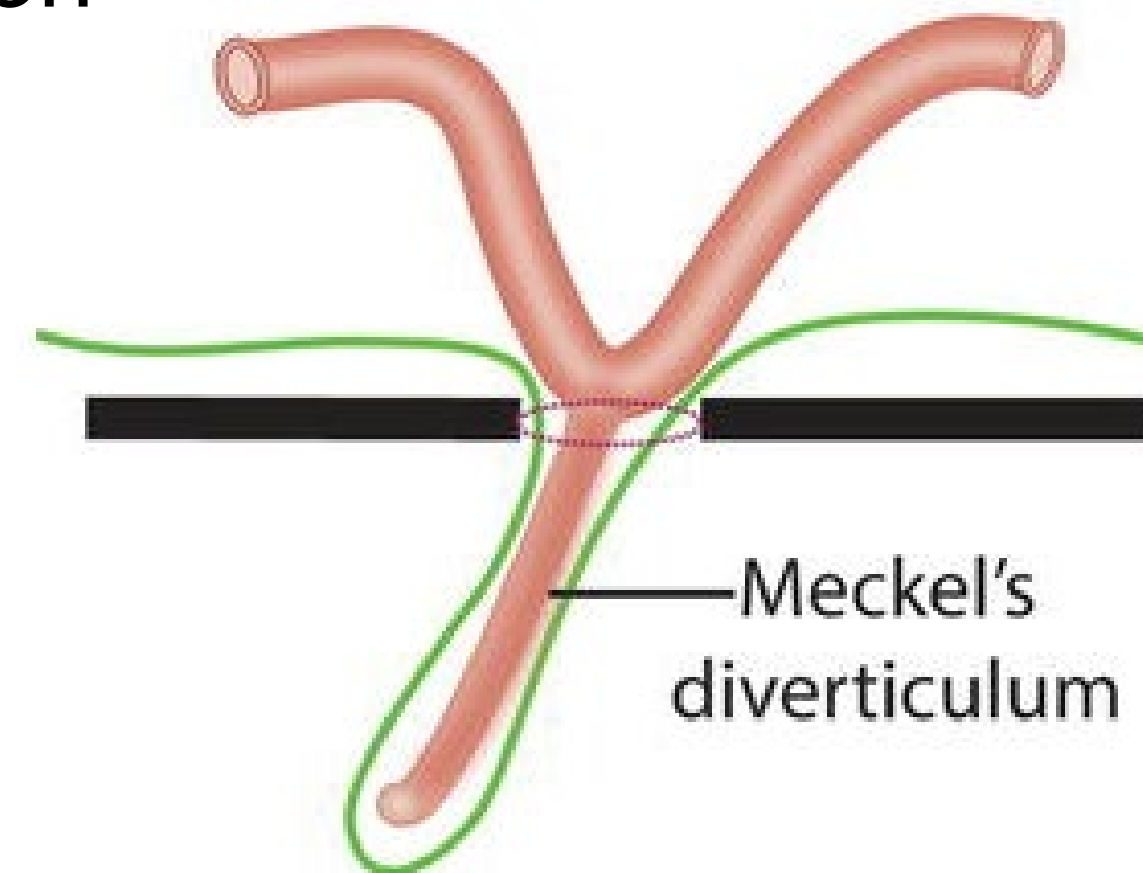
Pantaloon



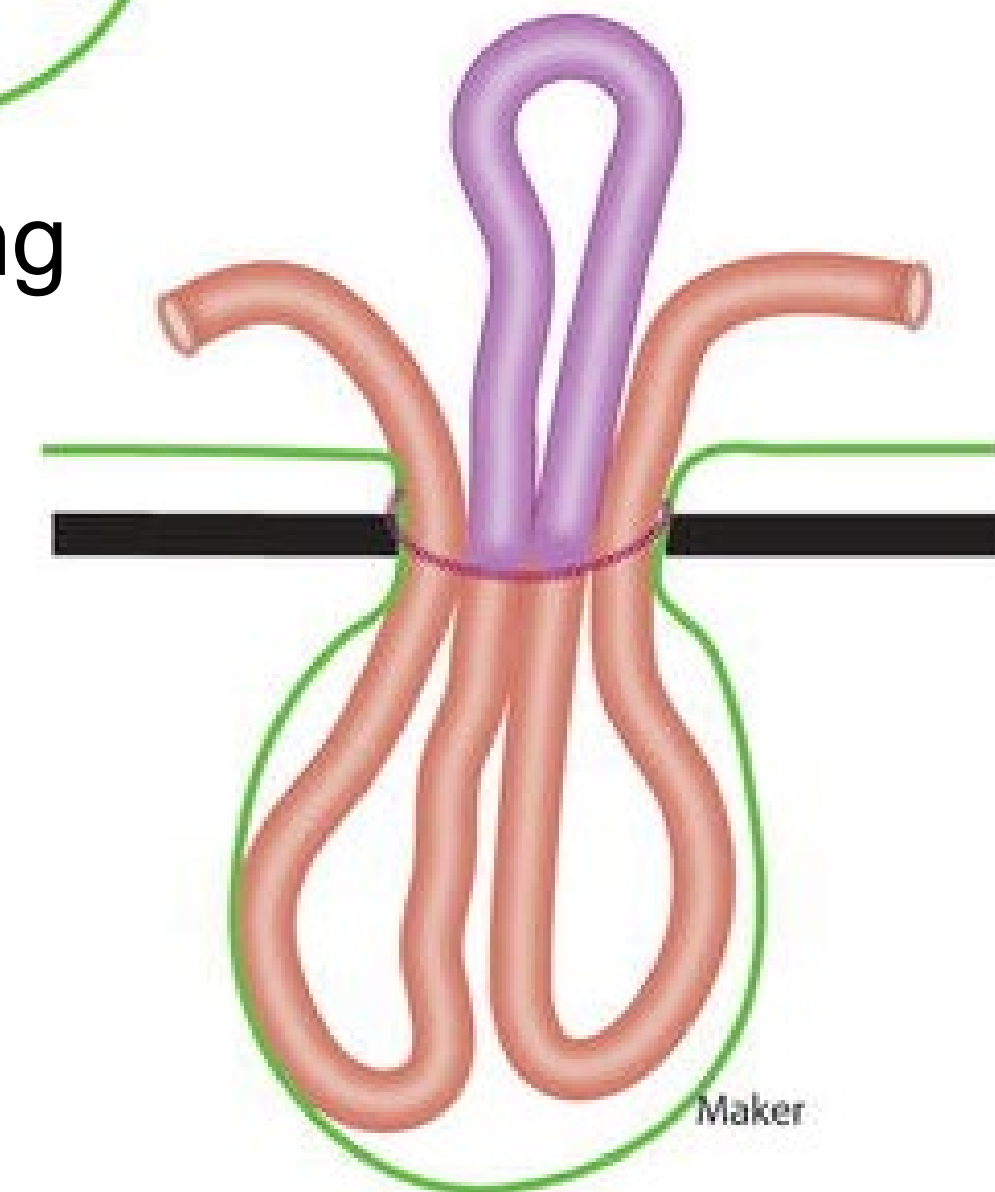
Sliding



Amyland

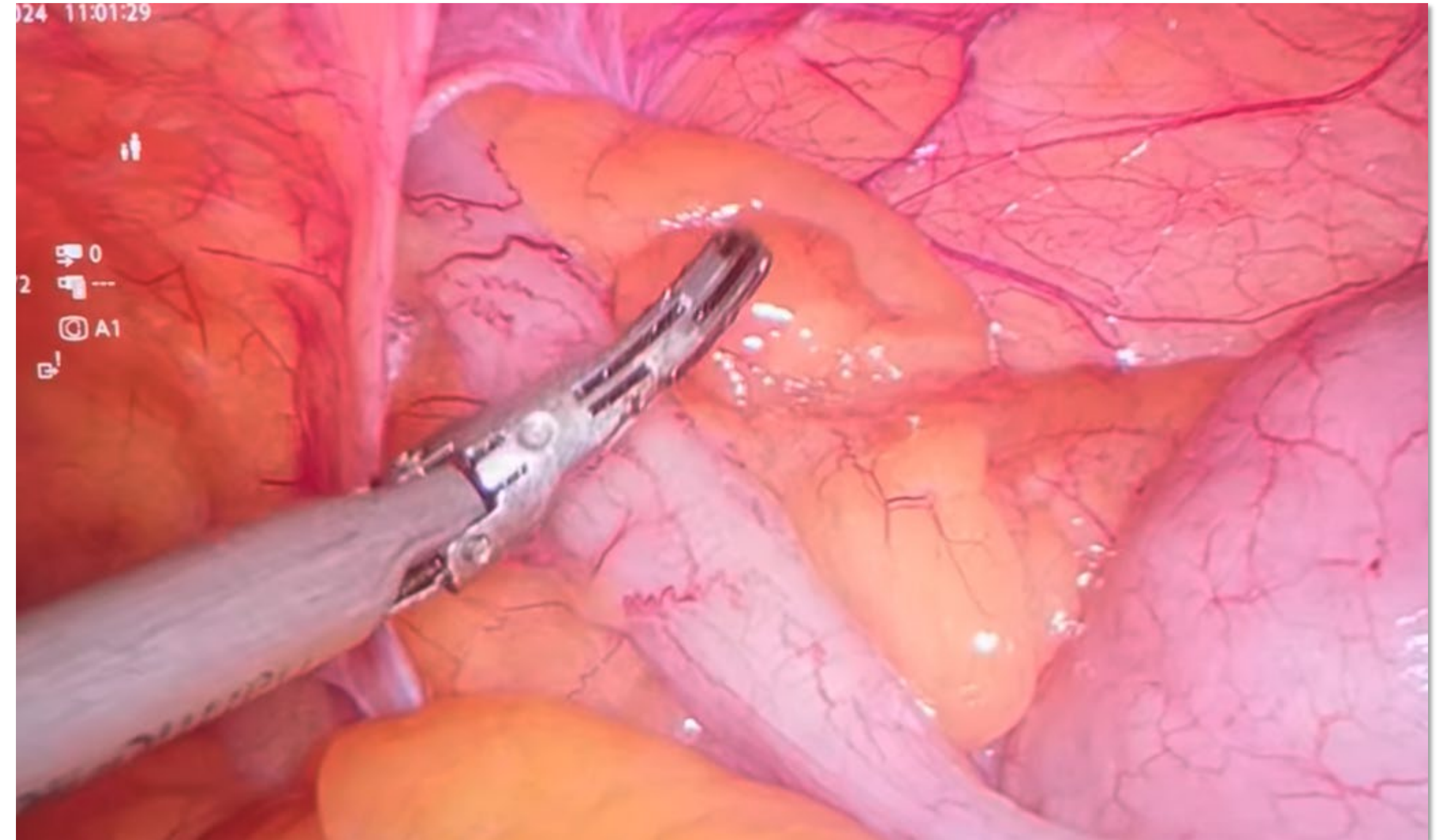
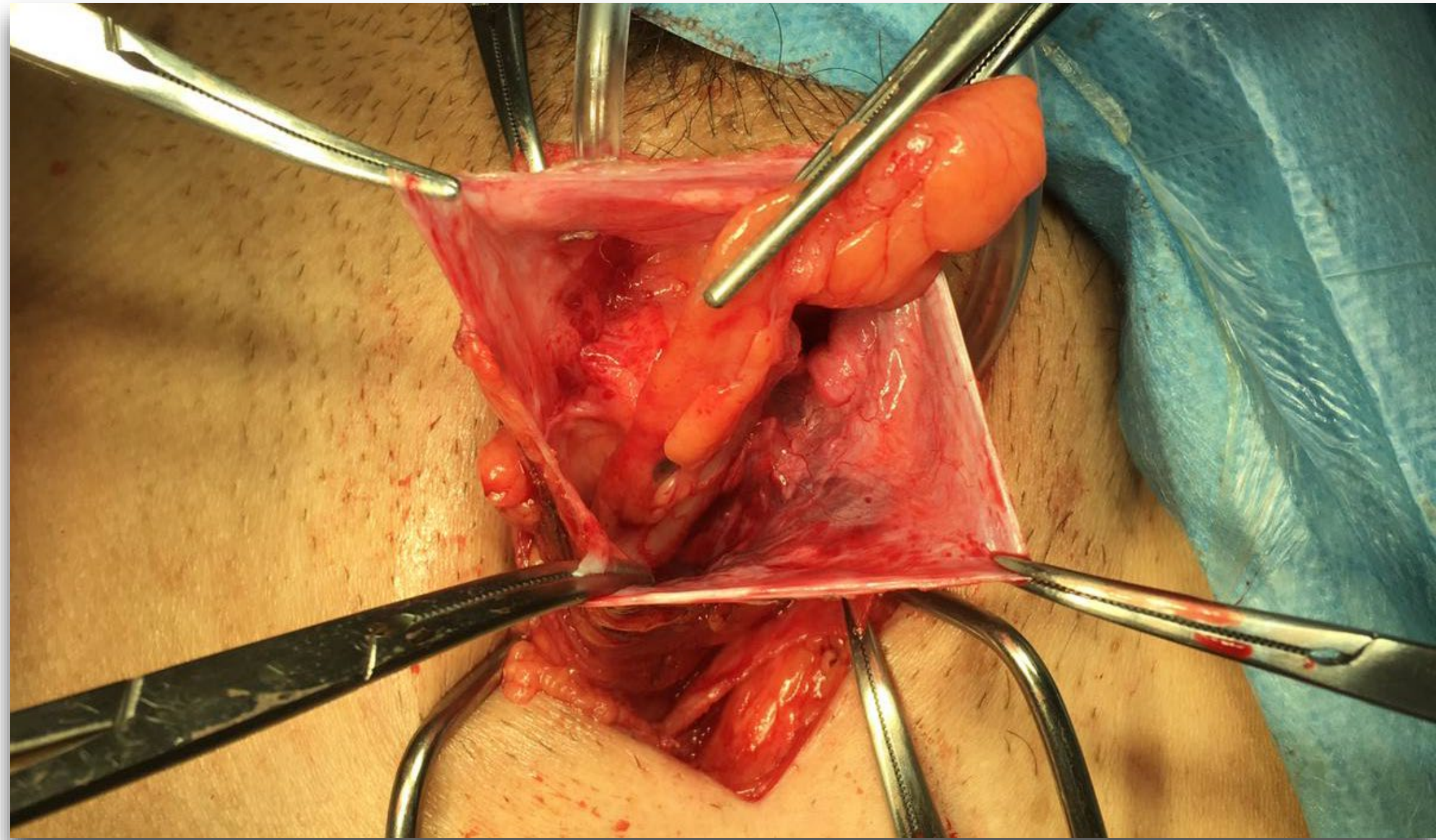


Littre



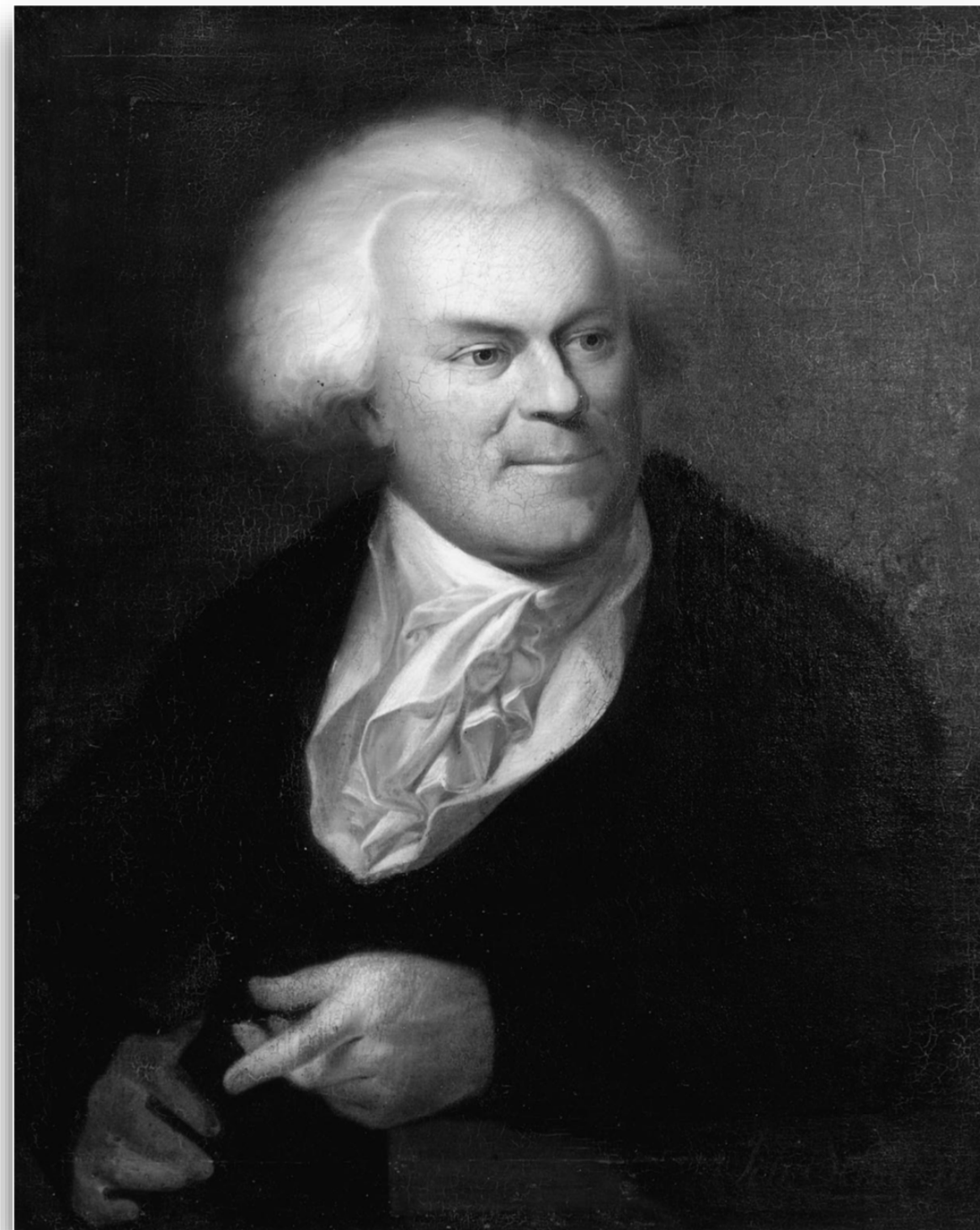
Maydl

Amyand's Hernia



Claudius Amyand, was a French surgeon who performed the first successful appendectomy in 1735, on an 11-year-old boy who presented with an inflamed, perforated appendix in his inguinal hernia sac.

Richter's Hernia



August Gottlob Richter (1742-1812)

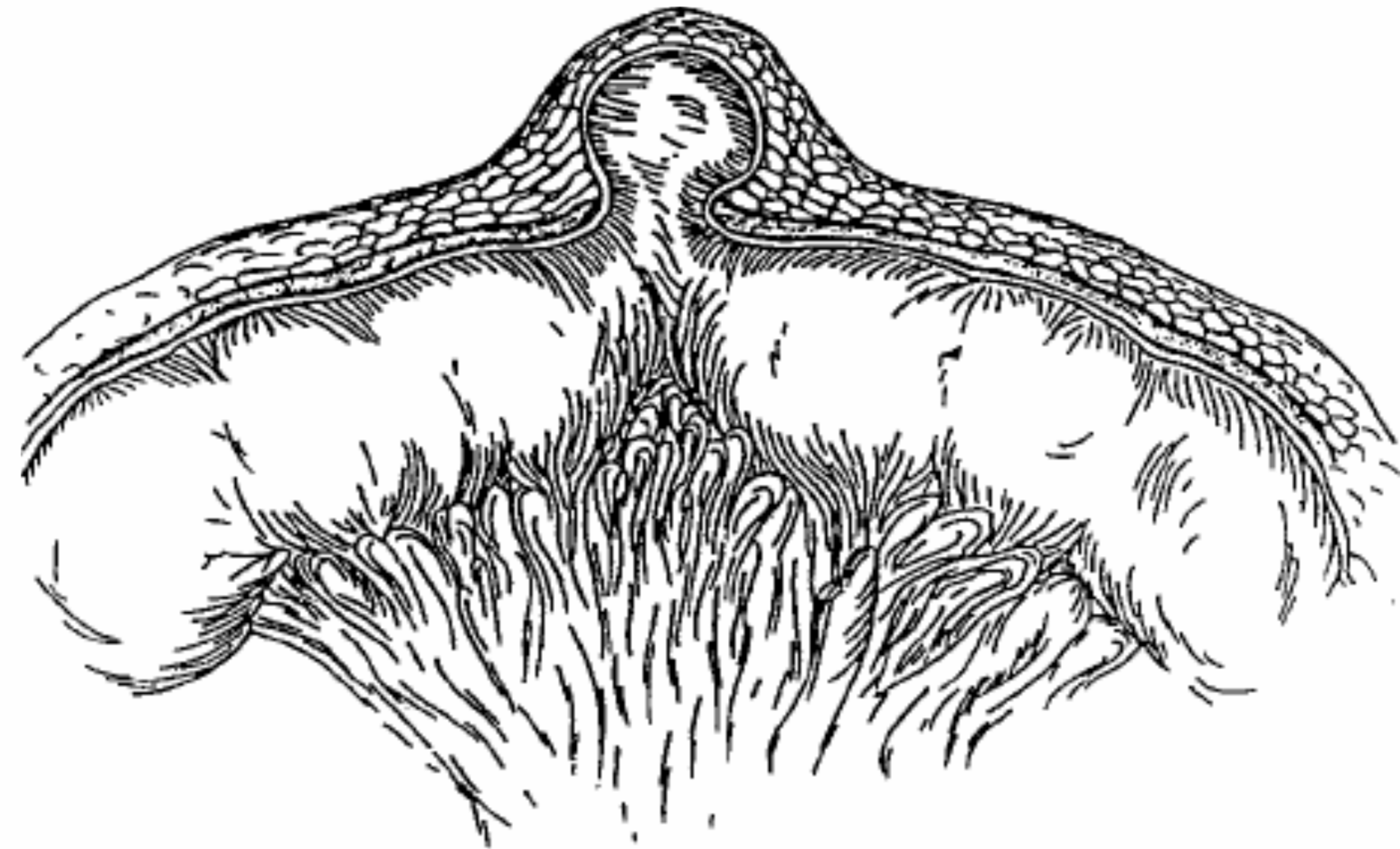


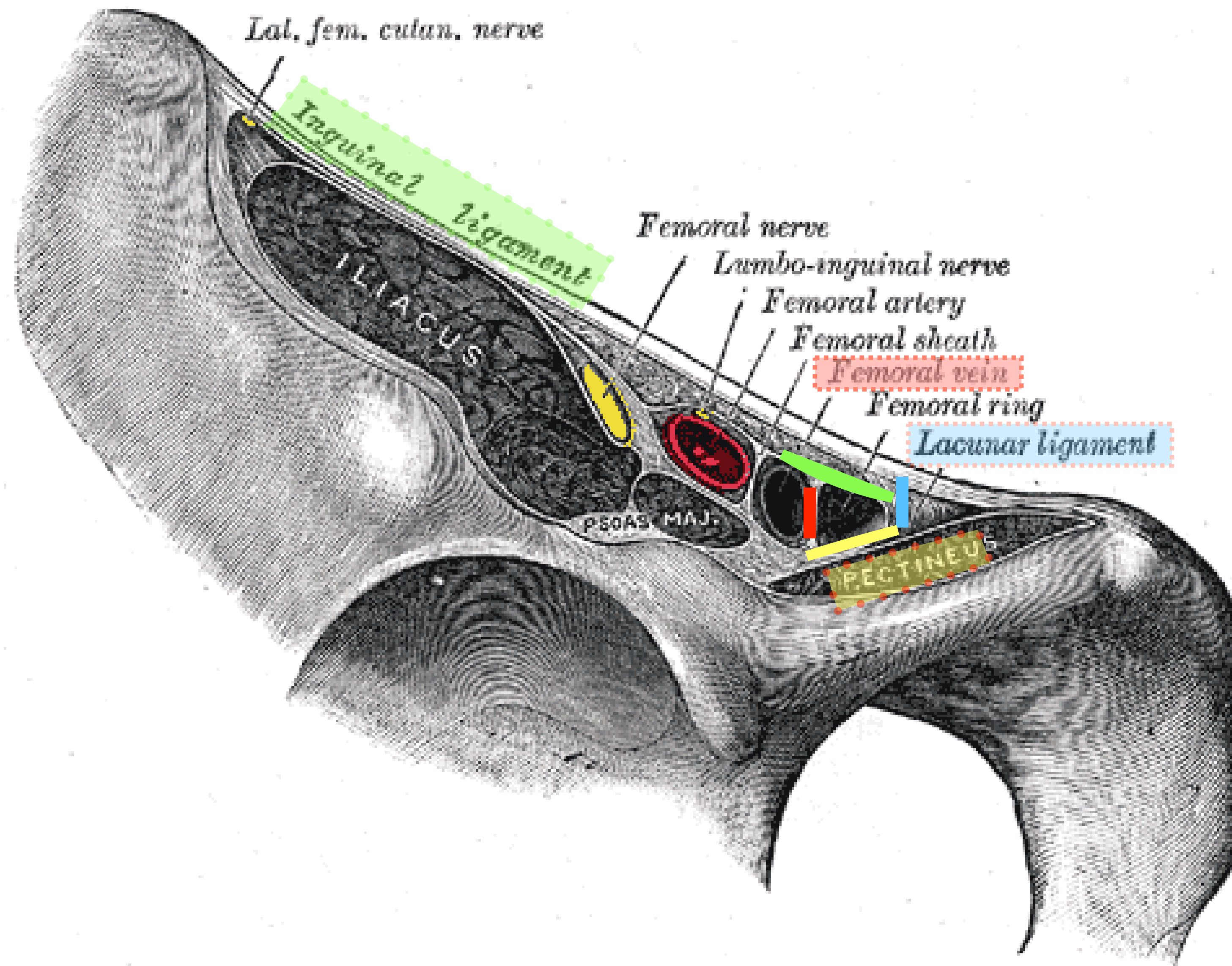
Figure 1. Partial entrapment of bowel wall with preservation of luminal continuity.

First described in 1778 by August Gottlob Richter who presented it as the small rupture

Femoral Hernia

- Less common than inguinal hernia. 2-8% of all adult groin hernias
- More common in females
- Higher chance of strangulation due to tight opening

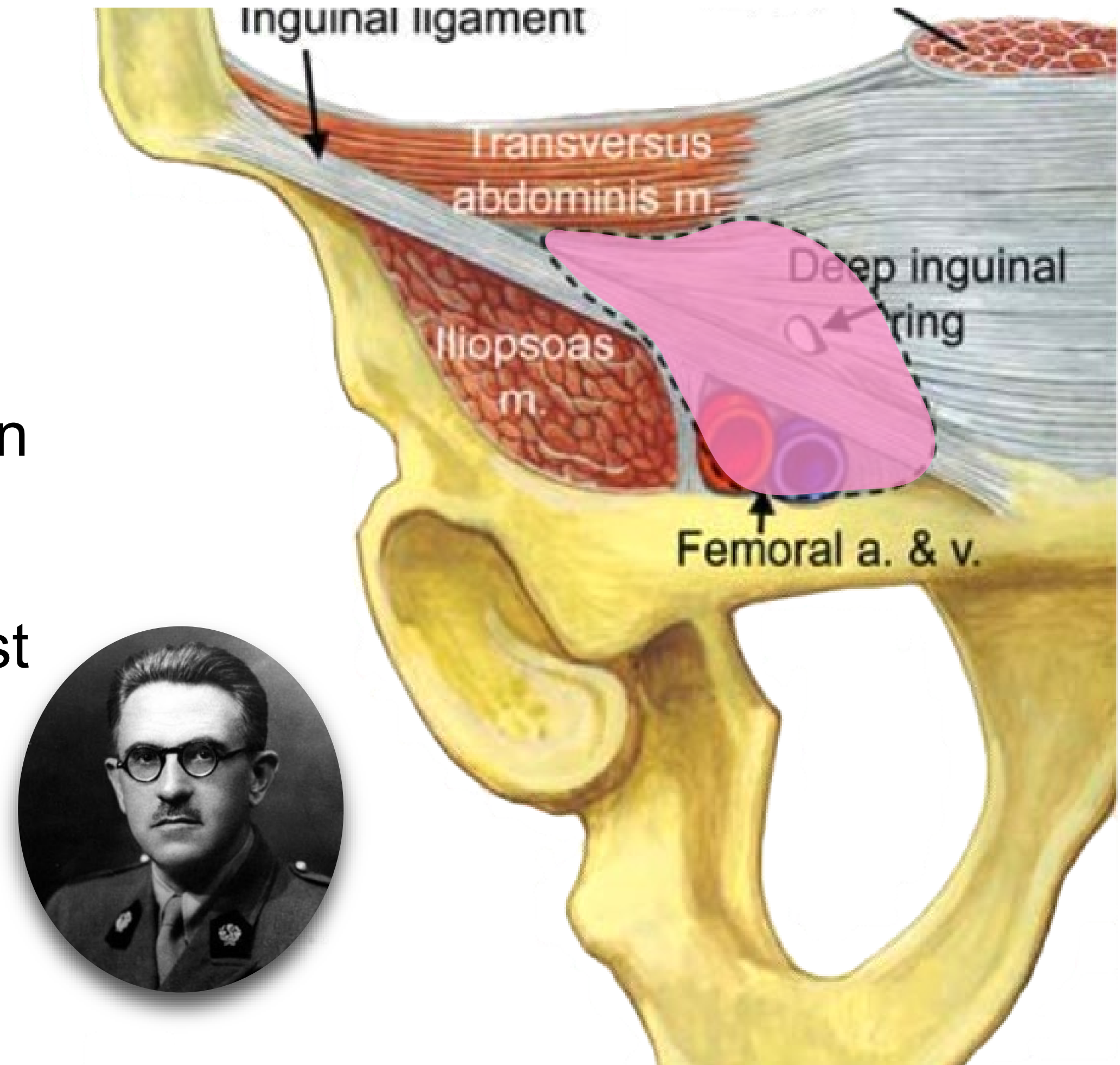
Sorelli, Paolo G, Nabil S El-Masry, and William V Garrett. "Open Femoral Hernia Repair: One Skin Incision for All.." *World Journal of Emergency Surgery : WJES* 4, no. 1 (November 30, 2009): 44. doi:10.1186/1749-7922-4-44.



"Gray546" by Henry Vandyke Carter - Henry Gray (1918) Anatomy of the Human Body (See "Book" section below)Bartleby.com: Gray's Anatomy, Plate 546. Licensed under Public Domain via Wikimedia Commons - <http://commons.wikimedia.org/wiki/File:Gray546.png#/media/File:Gray546.png>

Myopectineal Orifice “MPO”

- First described by Dr. Henri Fruchaud in 1956
- Area of weakness in the pelvic region.
- Normal congenital/anatomical gaps occur in this area
- Not reinforced by muscle layers like the rest of the abdominal wall
- More prone to repetitive increases in intra-abdominal pressure leading to progressive bulging, weakness and hernia formation.

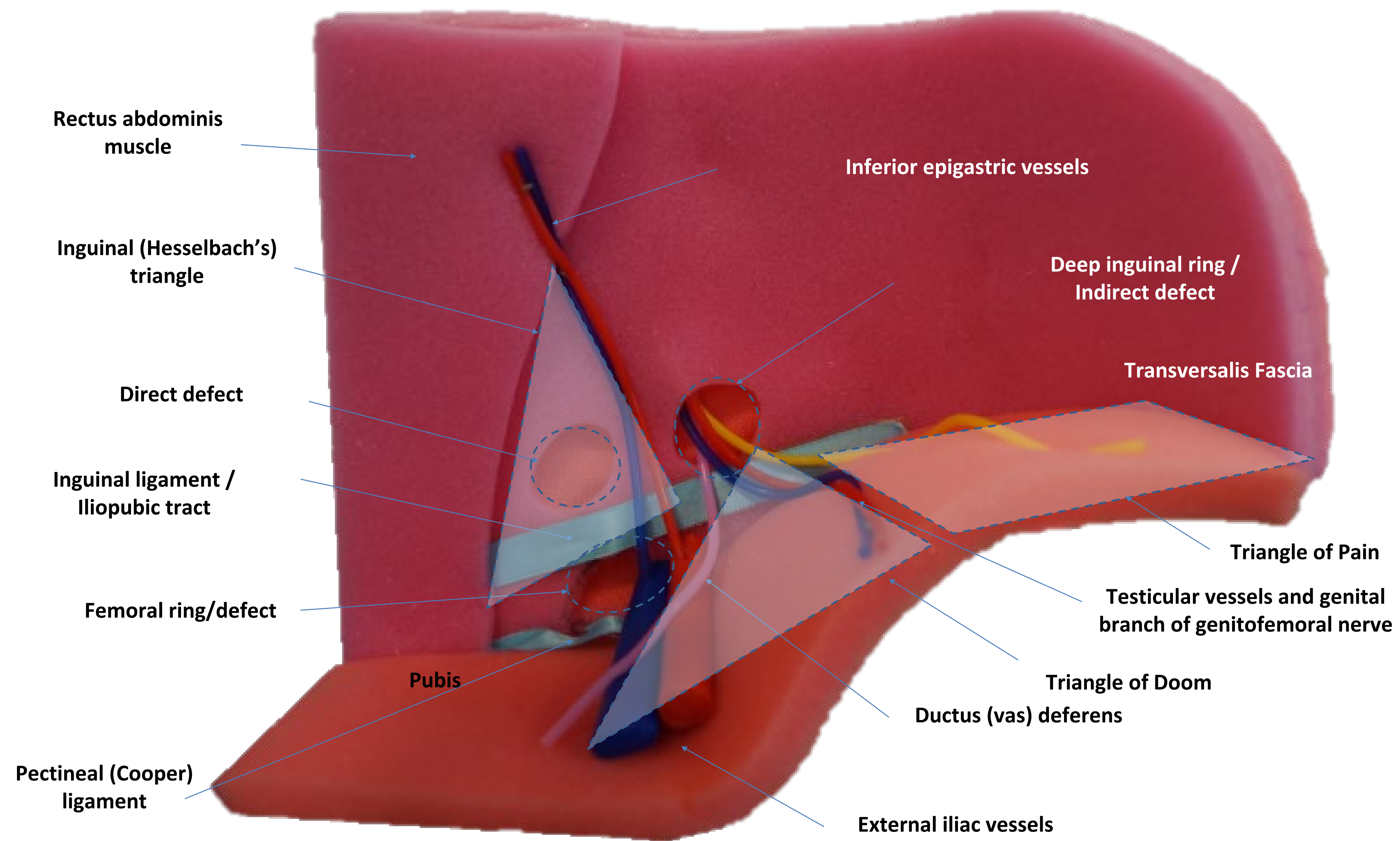


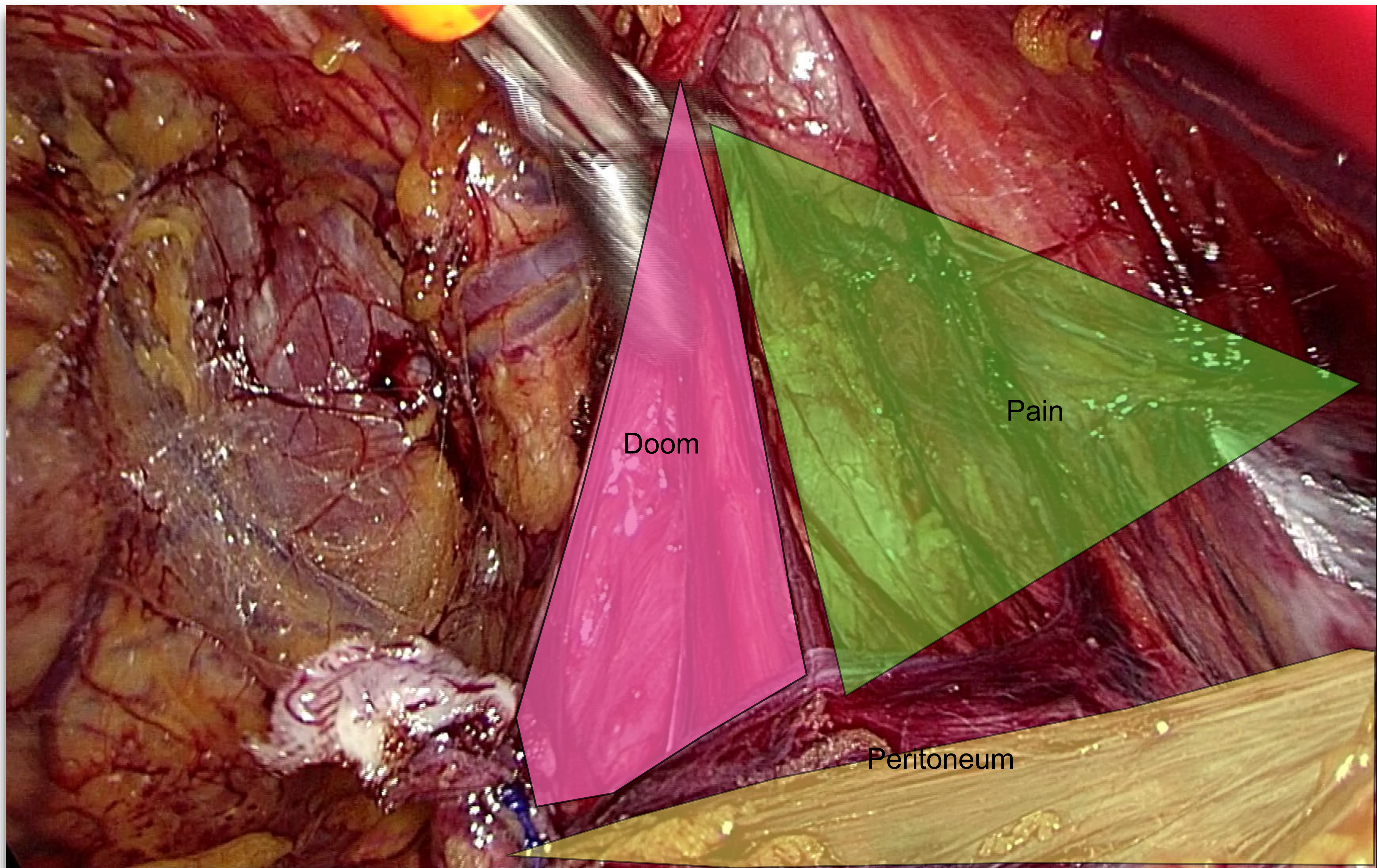
Fruchaud H (1956) Anatomie chirurgicale des hernies de l'aîne. Doin, Paris, pp 299–303 and pp 336–342

Wolloscheck, T, and M A Konerding. "Dimensions of the Myopectineal Orifice: a Human Cadaver Study." *Hernia* 13, no. 6 (September 11, 2009): 639–42. doi:10.1007/s10029-009-0559-1.

Where? Borders?

- **Above:** arching fibres of the internal oblique and transversus abdominus muscle
- **Medially:** by the rectus abdominus muscle and its fascial rectus sheath
- **Inferiorly:** by the Coopers ligament
- **Laterally:** by the Ileopsoas muscle

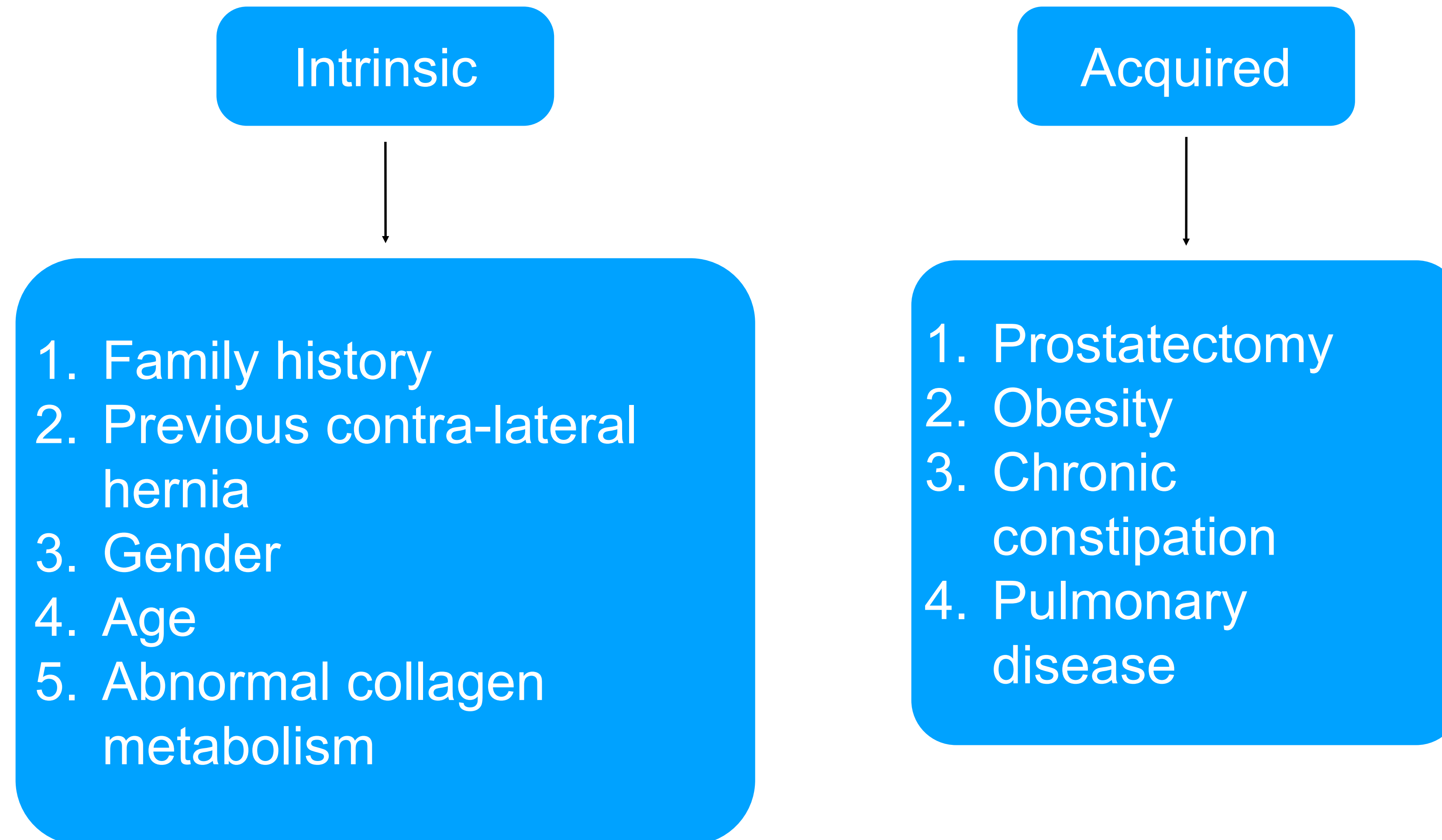




Typical history

- **Reducible** groin mass: Hernia
- **Painful** scrotal swelling: epididymo-orchitis/torsion
- **Painless** scrotal enlargement - Hydrocele/Testicular tumor
- Fullness/bag of **worms**: Varicocele

What are the risk factors for IH?



Diagnosis

- Physical examination alone
- Ultrasound
- CT
- Less commonly still MRI, herniography

Physical Examination

- Consent
- Exposure

Where do you start?



Stand



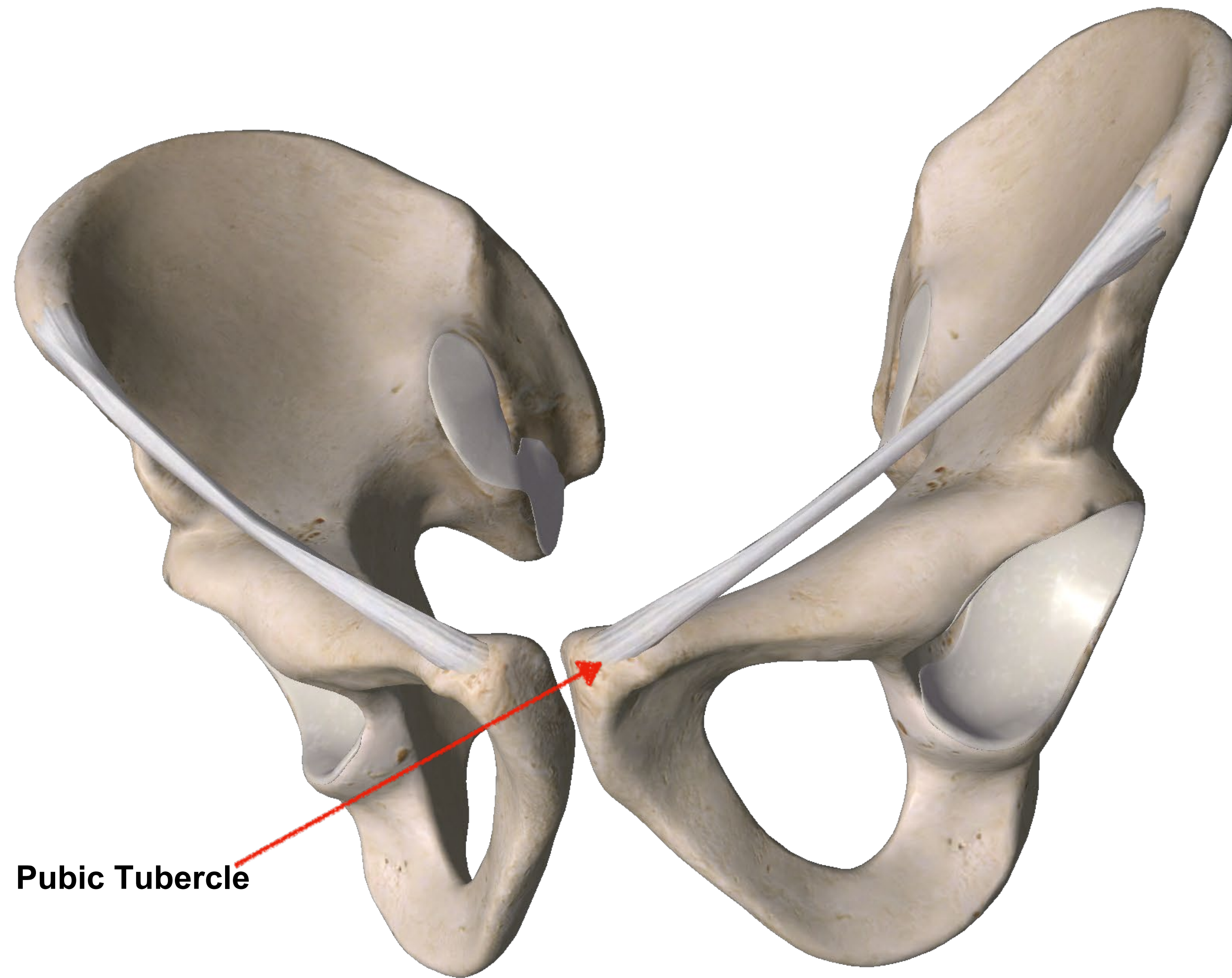
Supine

Standing

- **Inspect**
 - Describe the mass
 - Visible cough impulse?
 - Scars?
- **Palpate**
 - Describe the mass
 - Is it a scrotal mass or a groin mass extending to the scrotum? Can you get above it?
 - Palpable cough impulse? Unilateral? Bilateral?

Supine

- **Inspect**
- **Palpate**
 - Identify the pubic tubercle (PT) and the ASIS
 - Relationship with the PT?
 - Reducible?
 - Occlusion test
 - Abdominal mass?
 - Testis?

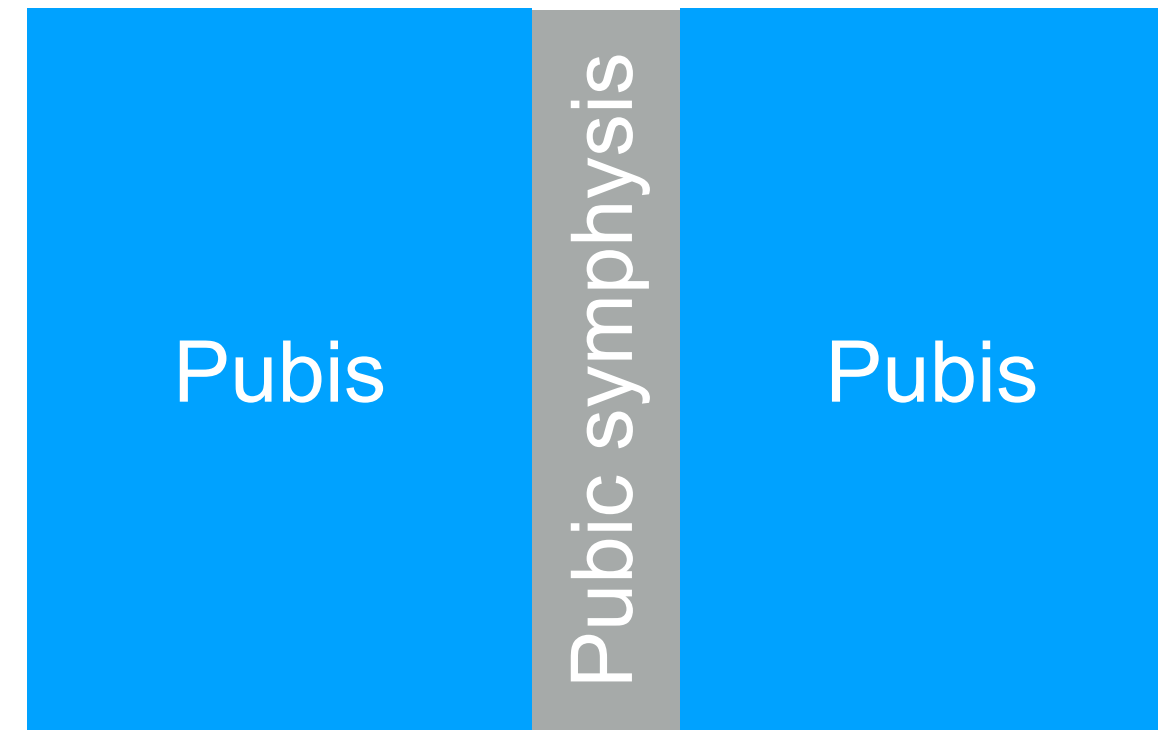
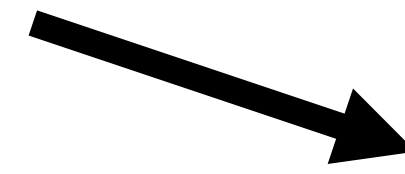


Pubic Tubercle

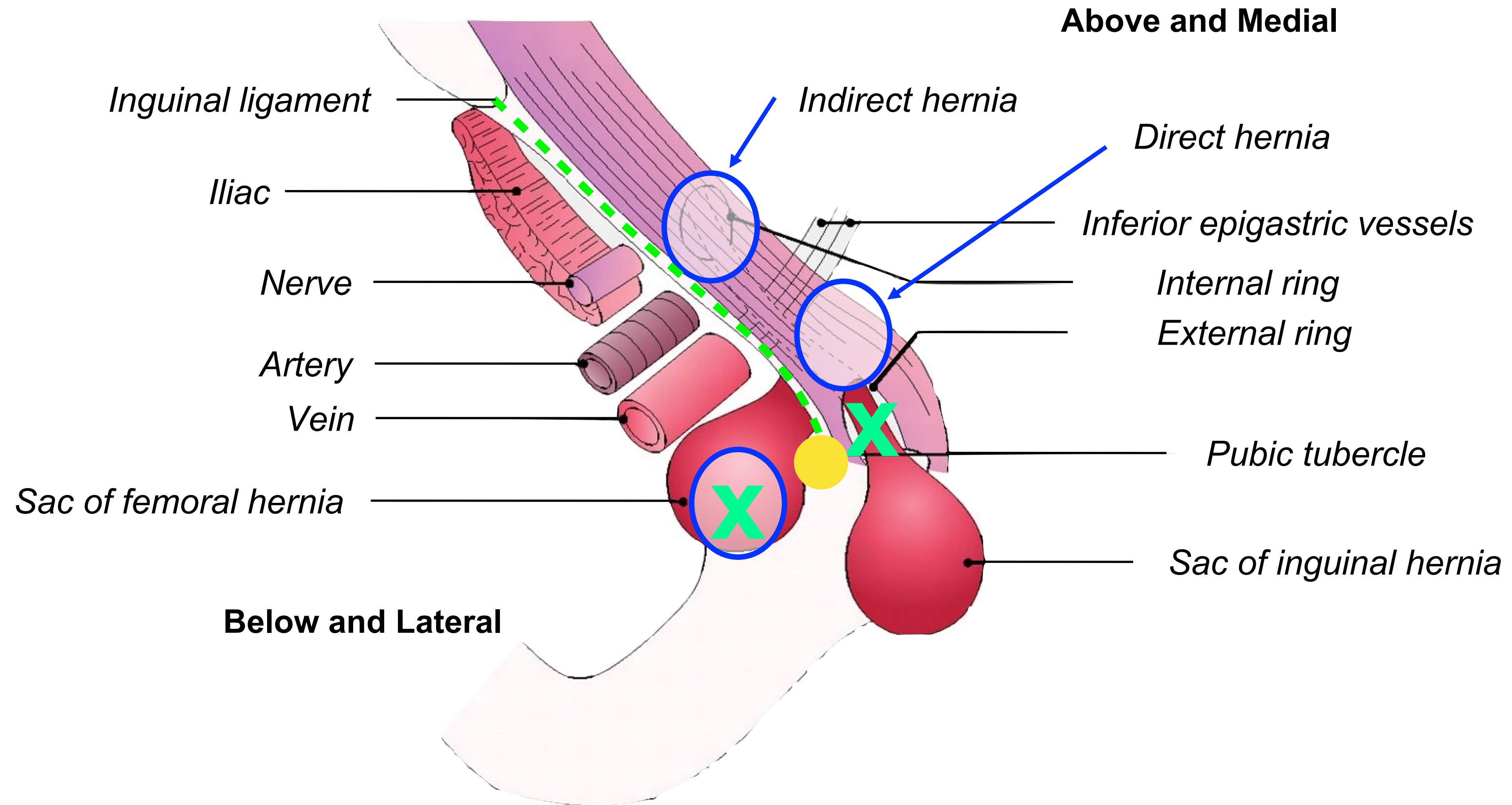
Image from 3D4Medical's Essential Anatomy 5 application.

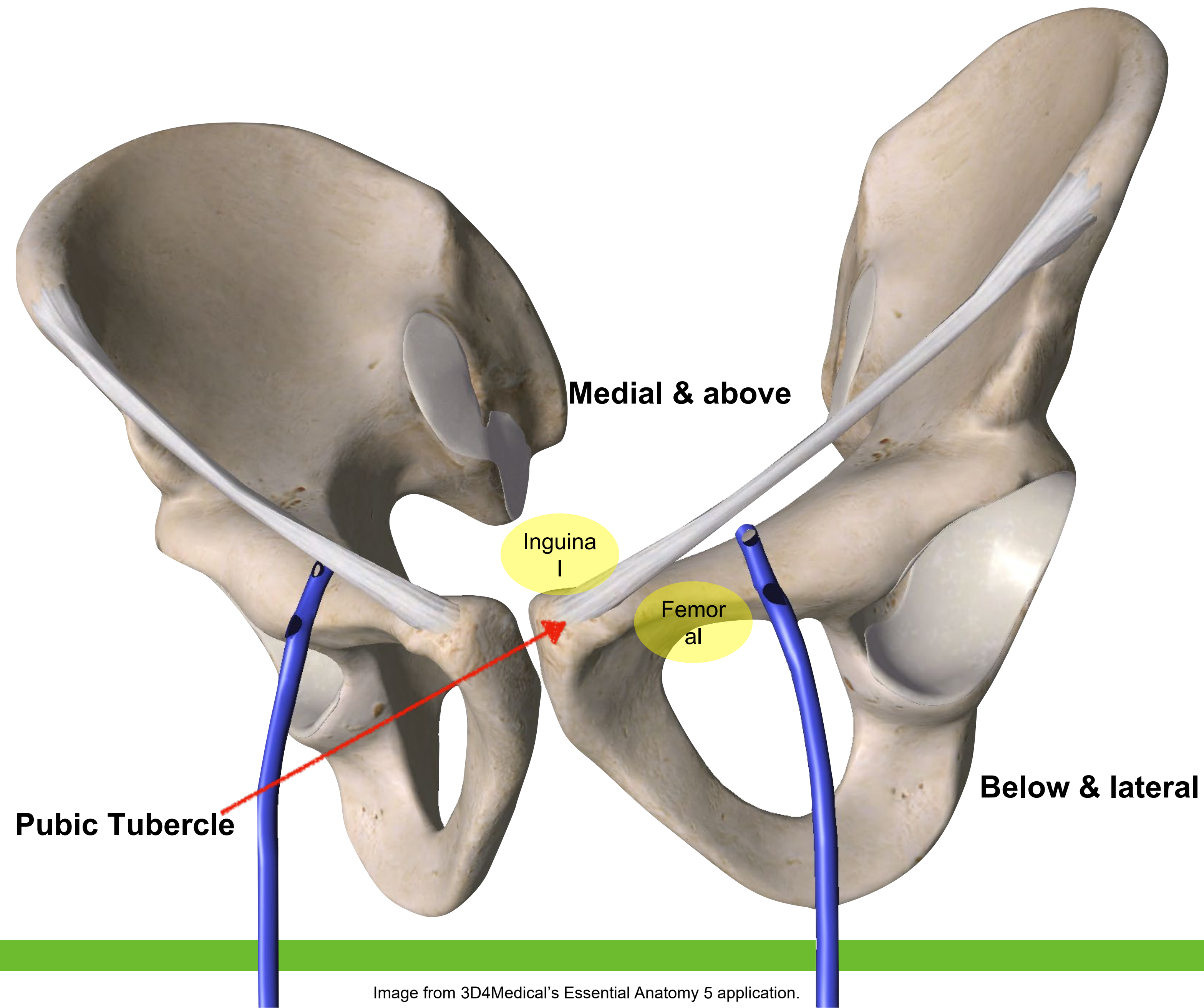


Pubic Tubercle



Anatomy





Medial & above

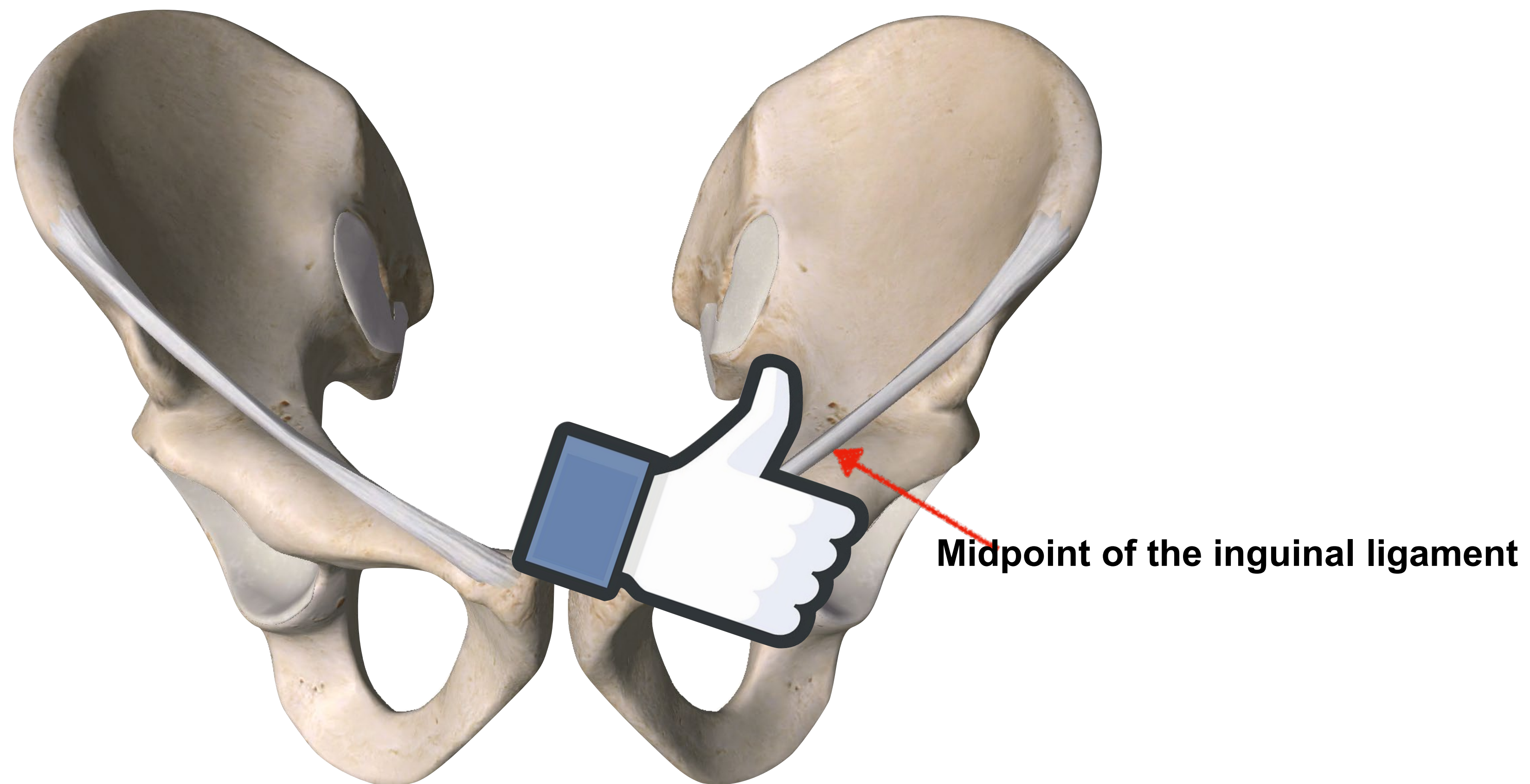
Inguina
I

Femor
al

Below & lateral

Pubic Tubercle

Occlusion Test



Occlusion Test

- Mid-inguinal point
 - midpoint between the ASIS and the **PS**. Marks the point where the femoral pulse can be palpated
- Mid-point of the inguinal ligament
 - midpoint between the ASIS and the **PT**. Marks where the deep inguinal ring lies.

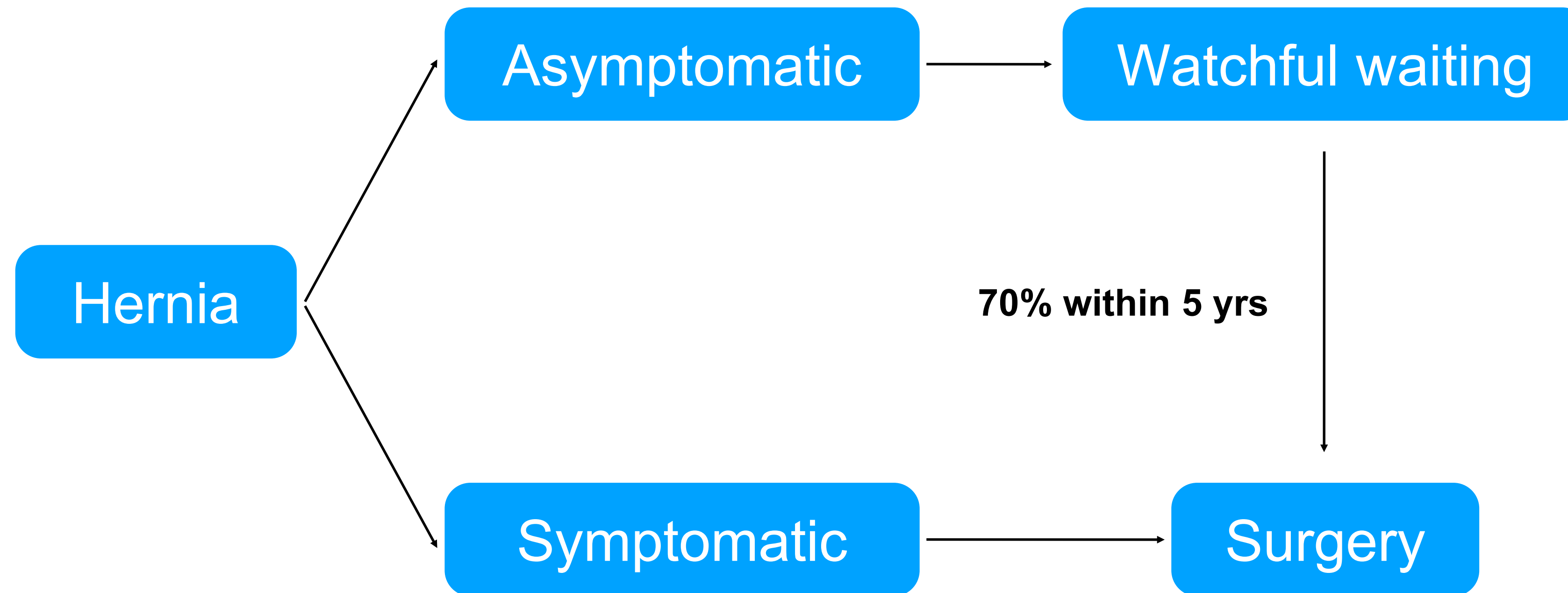
Occlusion Test

- Direct inguinal hernia
 - Accuracy of 35%
- Indirect inguinal hernia
 - Accuracy of 86%

Tromp, Wouter G, Baukje van den Heuvel, and Boudewijn J Dwars. "A New Accurate Method of Physical Examination for Differentiation of Inguinal Hernia Types." Surgical Endoscopy 28, no. 5 (January 8, 2014): 1460–64. doi:10.1007/s00464-013-3359-8.

- Examination of the Abdomen
 - Mass?
- Examination of the Scrotum
 - Both testis present? Undescended testis?

Treatment



Hernia Repair

Herniotomy

Excision of the hernia sac after reduction of contents

Herniorrhaphy

Herniotomy + Strengthening of the posterior wall without mesh

Hernioplasty

Herniotomy + Strengthening of posterior wall with mesh

Treatment

Tissue Repair

Bassini

Shouldice

Mesh Repair

Open

Anterior

ie. Lichtenstein

Posterior

ie. Stoppa

Laparoscopic

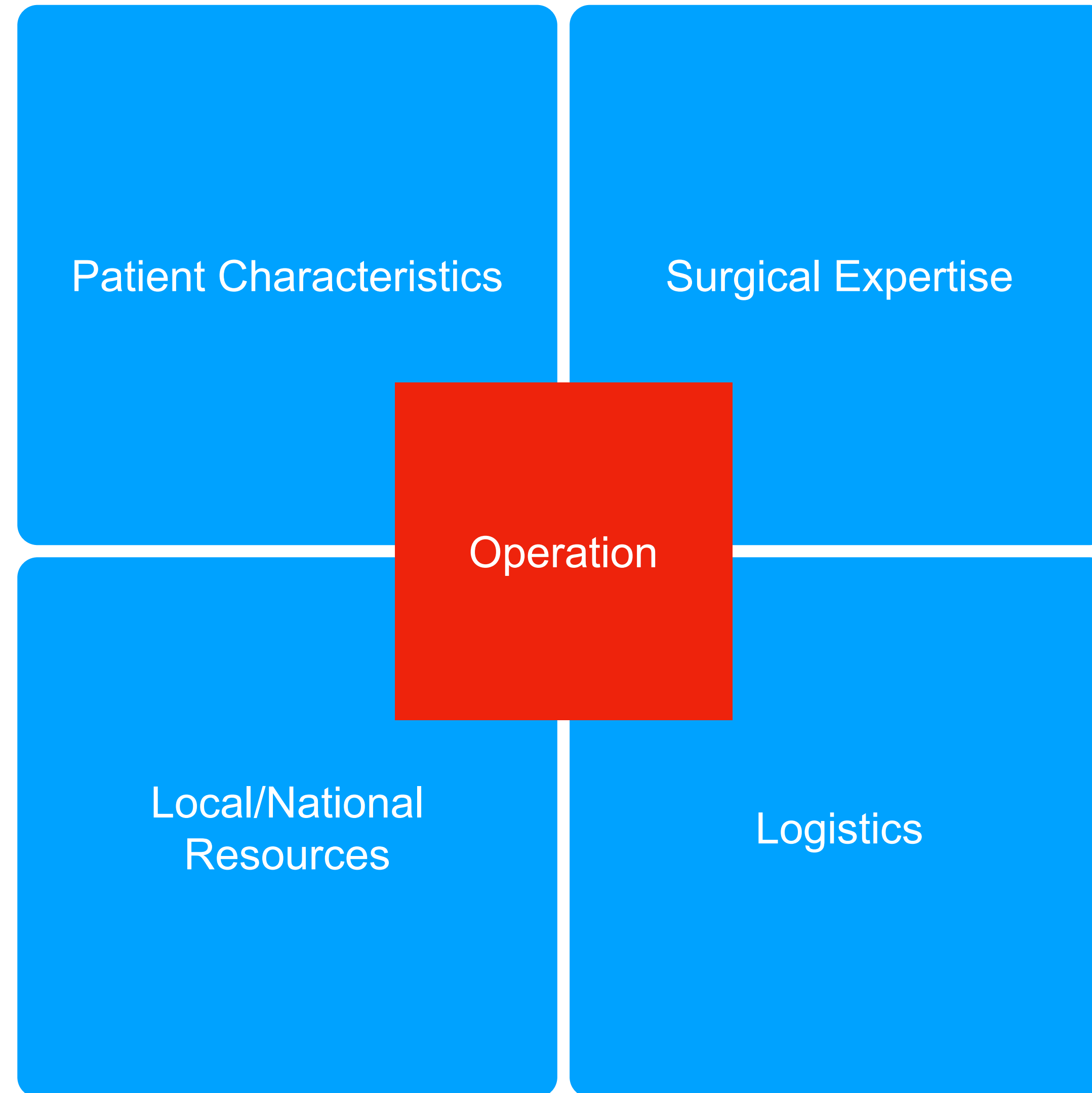
Posterior

TEP

TAPP

eTEP

Which approach?



Preoperative risk factors for recurrence

Poor surgical **technique**

Low surgical **volume**

Surgical **Inexperience**

Best operative technique?



Low risk of complications (pain and recurrence)



Easy to learn



Fast recovery



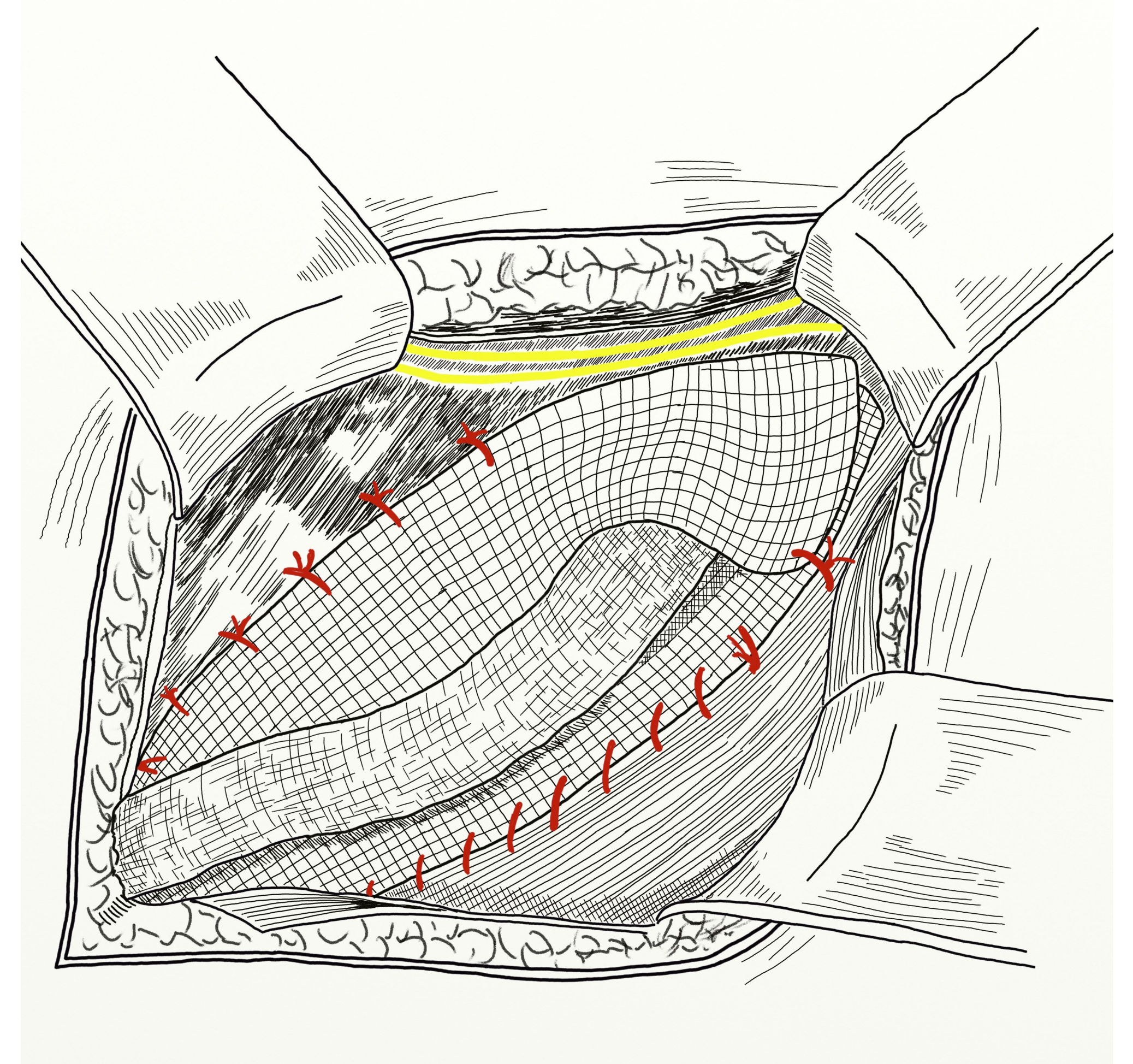
Reproducible result



Cost effective

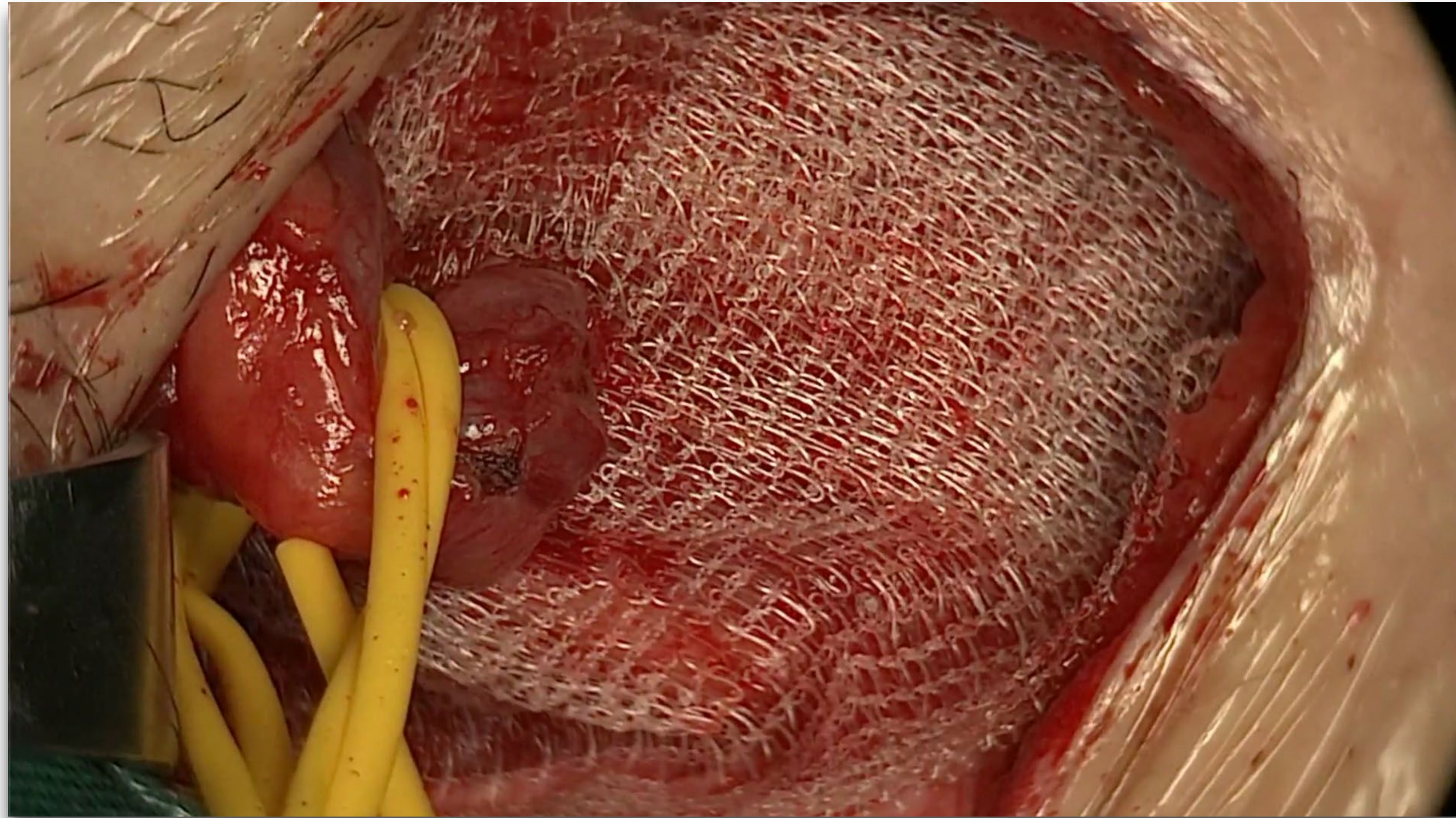
Lichtenstein Repair

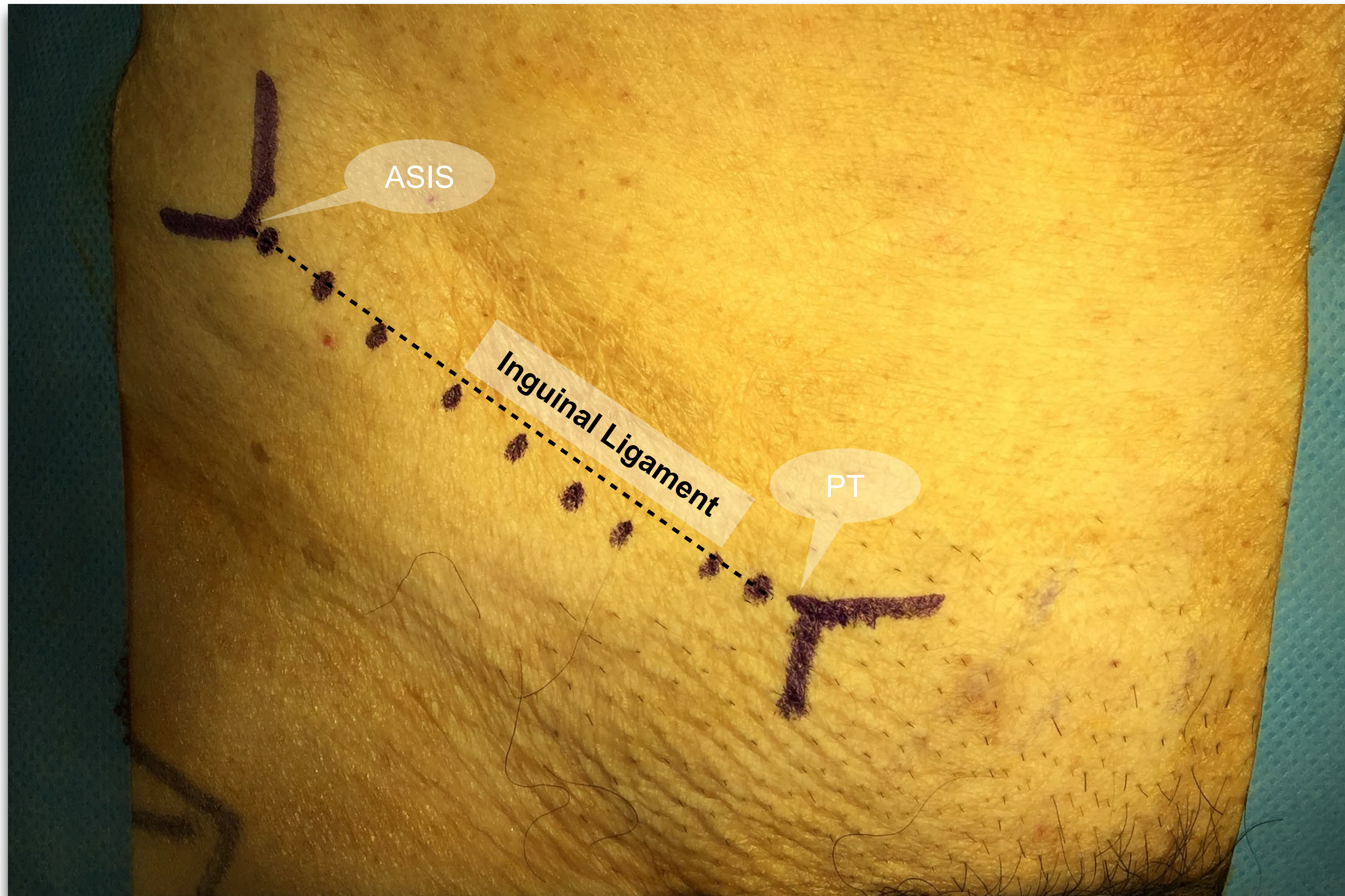
- First published in 1986¹
- Gold standard and the most common type of repair performed worldwide.² One of the most commonly performed operation worldwide.
- Easy to **learn, perform, teach, and effective.**²
- Recurrence rate of less than 1%²

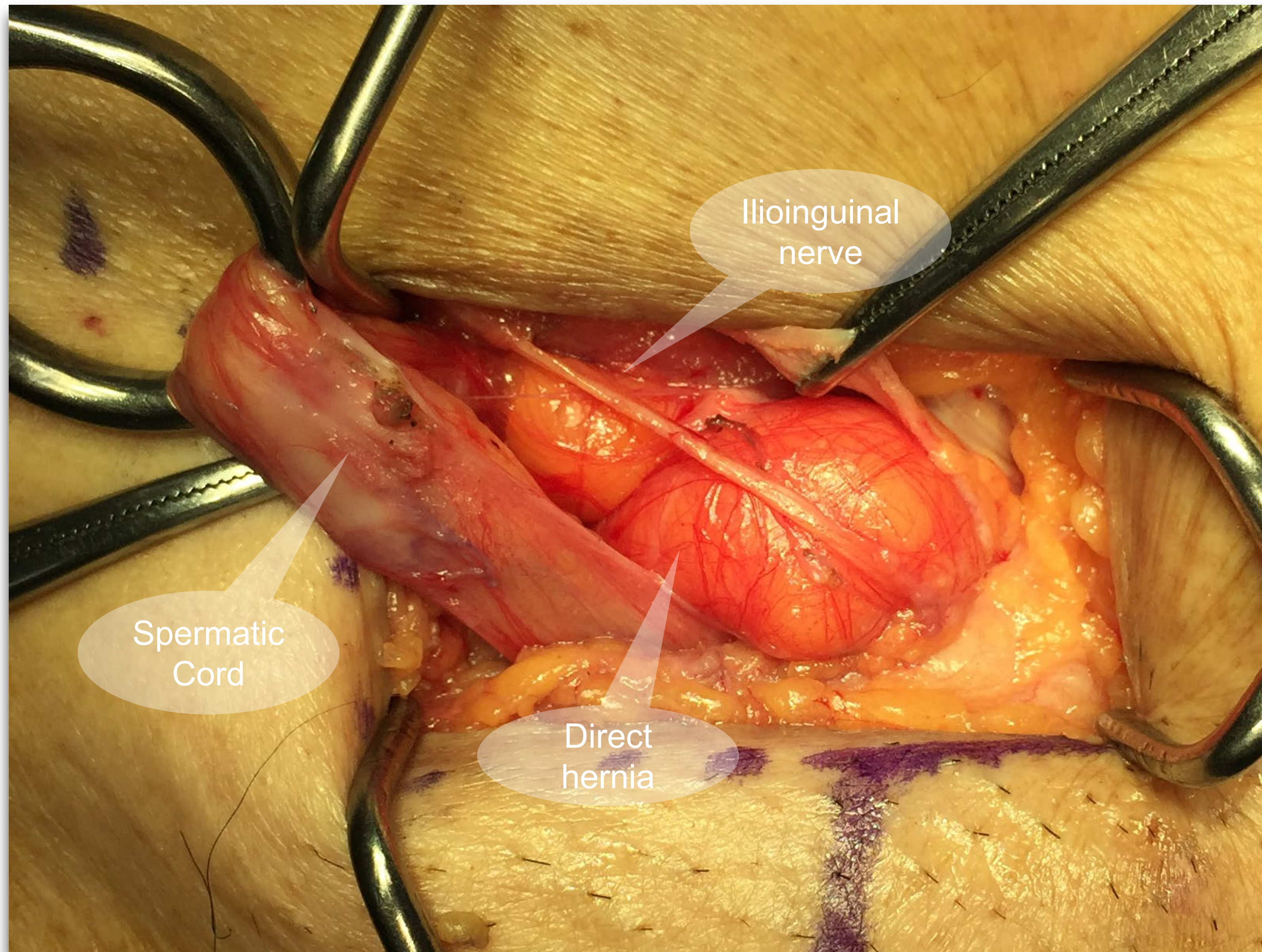


https://commons.wikimedia.org/wiki/File:Tension-free_Operation_Lichtenstein.jpg#filelinks

Open Tensionless Mesh Repair



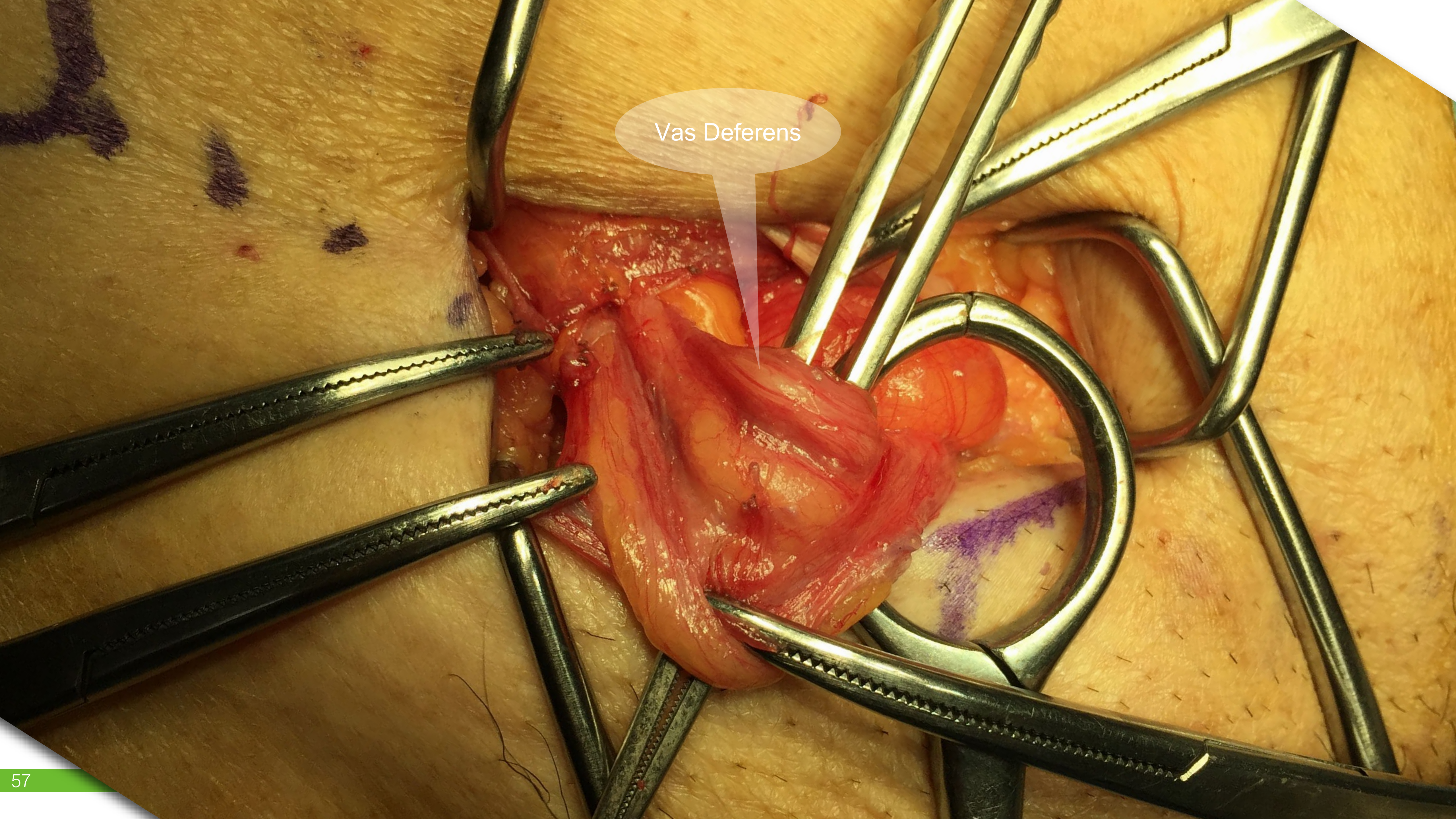




Spermatic
Cord

Ilioinguinal
nerve

Direct
hernia



Vas Deferens

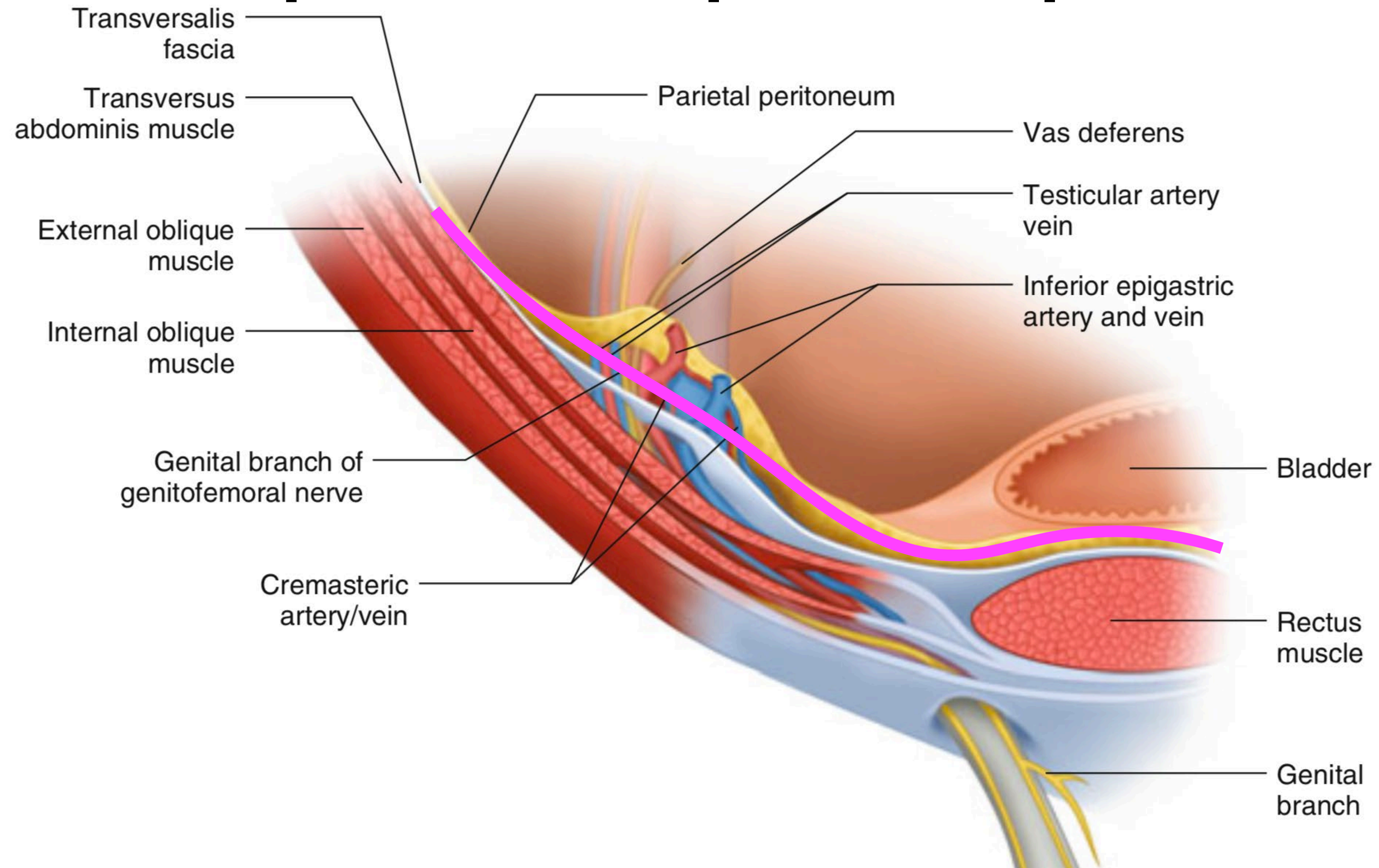
Shouldice repair

- Best tissue, non-mesh technique
- Difficult to learn.
- In the Shouldice Hospital, surgeons are only considered qualified after 300 cases.

Endo/Laparoscopic Repair

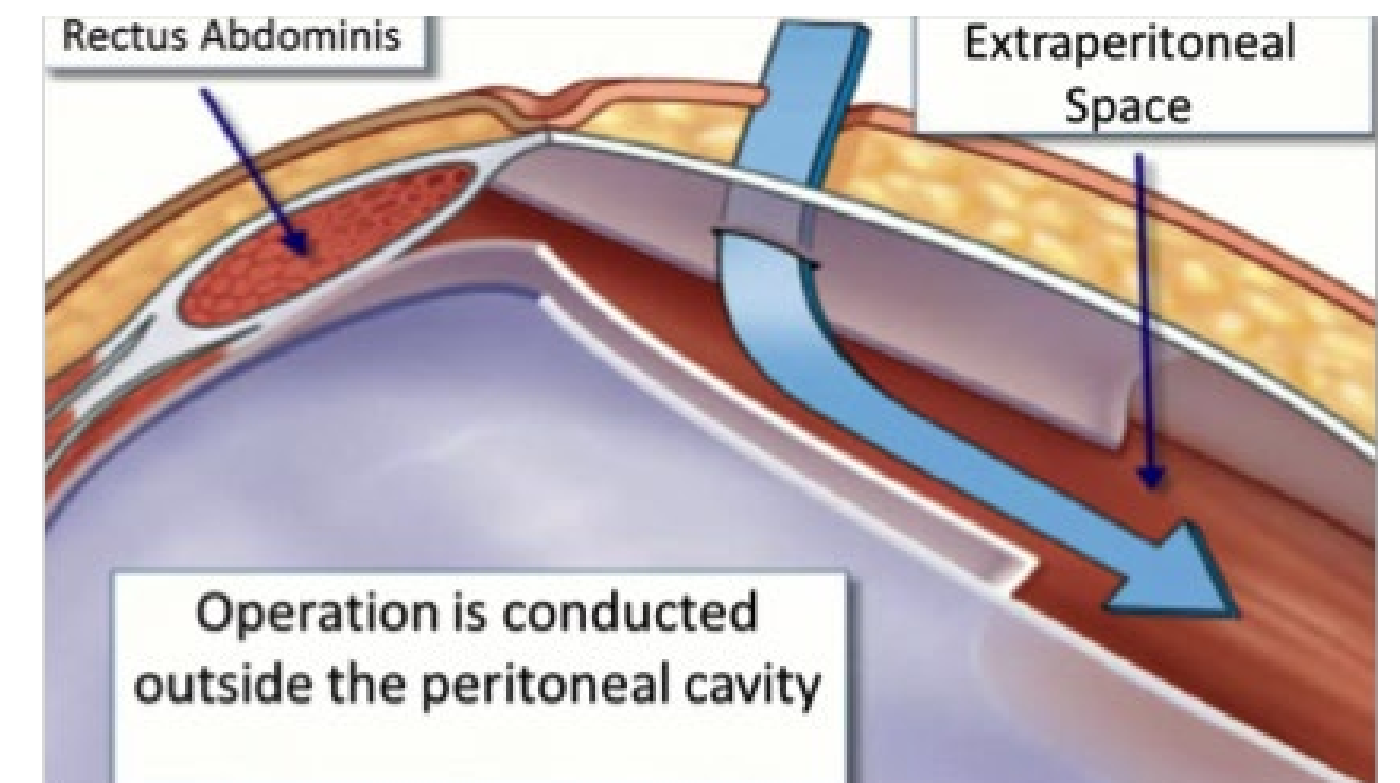
- Well accepted alternative to open repair
- Advantages in terms of less early postoperative pain, less chronic pain.
- European Hernia Society Guideline
 - Endoscopic repair for both unilateral and bilateral hernias
- Preferred approach when dealing with recurrent hernia after open repair

Laparoscopic Repair



Laparoscopic repair

- Total Extraperitoneal Repair (TEP)
 - Directly dissect into the pre peritoneal plane without perforating the peritoneum
- Transabdominal Preperitoneal Repair (TAPP)
 - Entering the peritoneal cavity first (diagnostic laparoscopy) then incising the peritoneum to enter the pre peritoneal plane



TAPP or TEP?

- TAPP and TEP have comparable outcomes
- choice depends on surgeon's skills, education and experience
- TEP has a longer learning curve than TAPP
- TAPP more visceral injuries
- TEP more vascular injuries

Females?

- Laparo-endoscopic recommended
- Decrease chronic pain risk
- Avoid missing a femoral hernia
- Pregnant women?
 - Watchful waiting -> self-limiting round ligament varicosities



Nyhus Classification

Classification designed for the posterior approach based on the size of the internal ring and the integrity of the posterior wall.

- Type 1 is an indirect hernia with a normal internal ring;
- Type 2 is an indirect hernia with an enlarged internal ring;
- Type 3a is a direct inguinal hernia;
- Type 3b is an indirect hernia causing posterior wall weakness;
- Type 3c is a femoral hernia;
- Type 4 represents all recurrent hernias.

EHS Groin Hernia Classification	Primary Recurrent				
	0	1	2	3	x
Lateral (Indirect)			x	Primary L2	
Medial (Direct)					
Femoral					

1: ≤ 1 finger

2: 1-2 fingers

3: ≥3 fingers



Chronic Groin Pain Inguinodynia

Primary groin pain: Groin pain not related to a prior surgery

Secondary groin pain: Groin pain that began after a surgical procedure (including hernia repairs and orthopedics surgery)

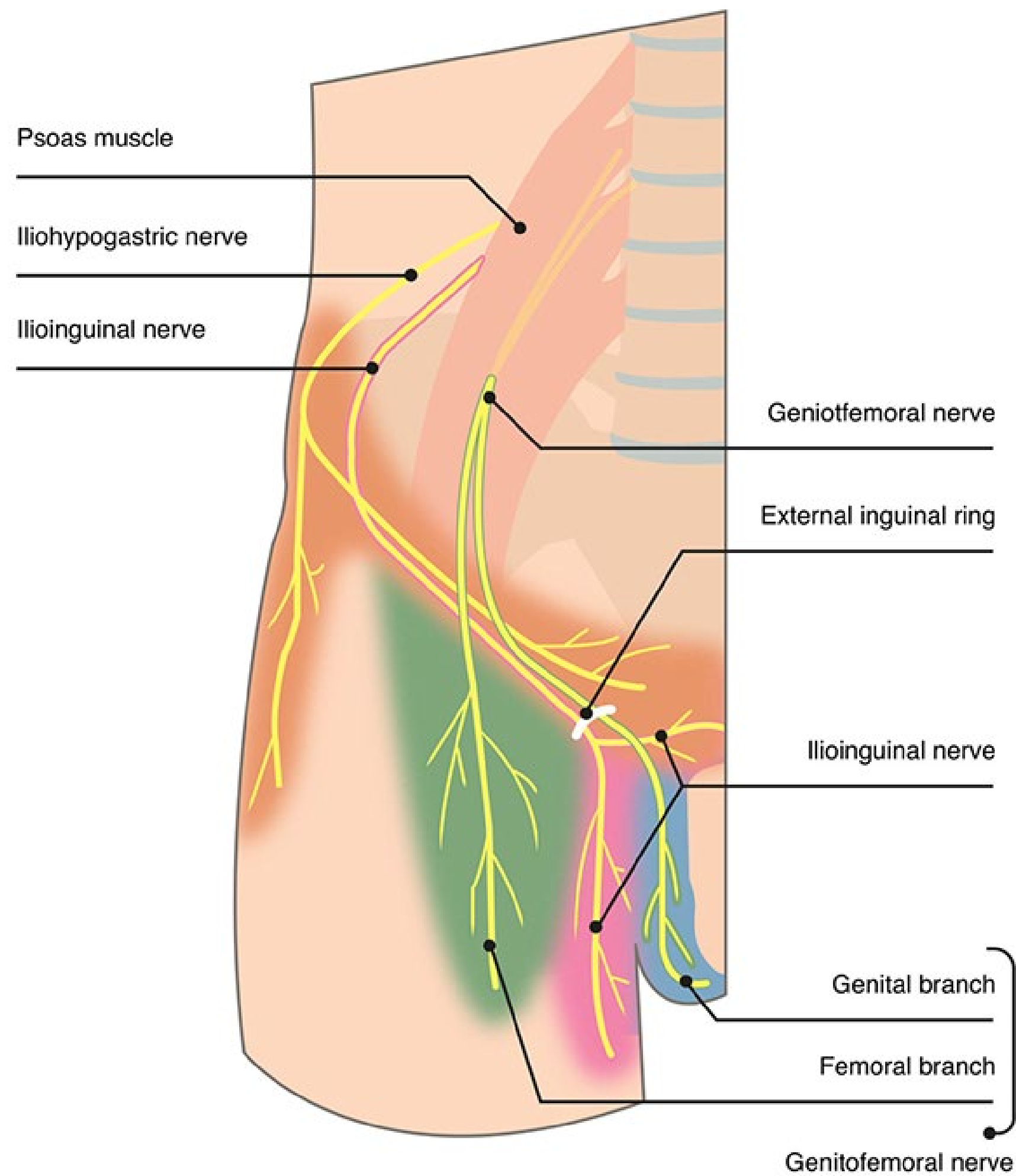
Chronic postoperative inguinal pain (CPIP)

- 10-12% of IH repair patients
- Bothersome moderate pain impacting daily activities lasting at least 3 months postoperatively
- Risk factors include: young age, female gender, high preoperative pain, early high postoperatively pain, recurrent hernia and open repair

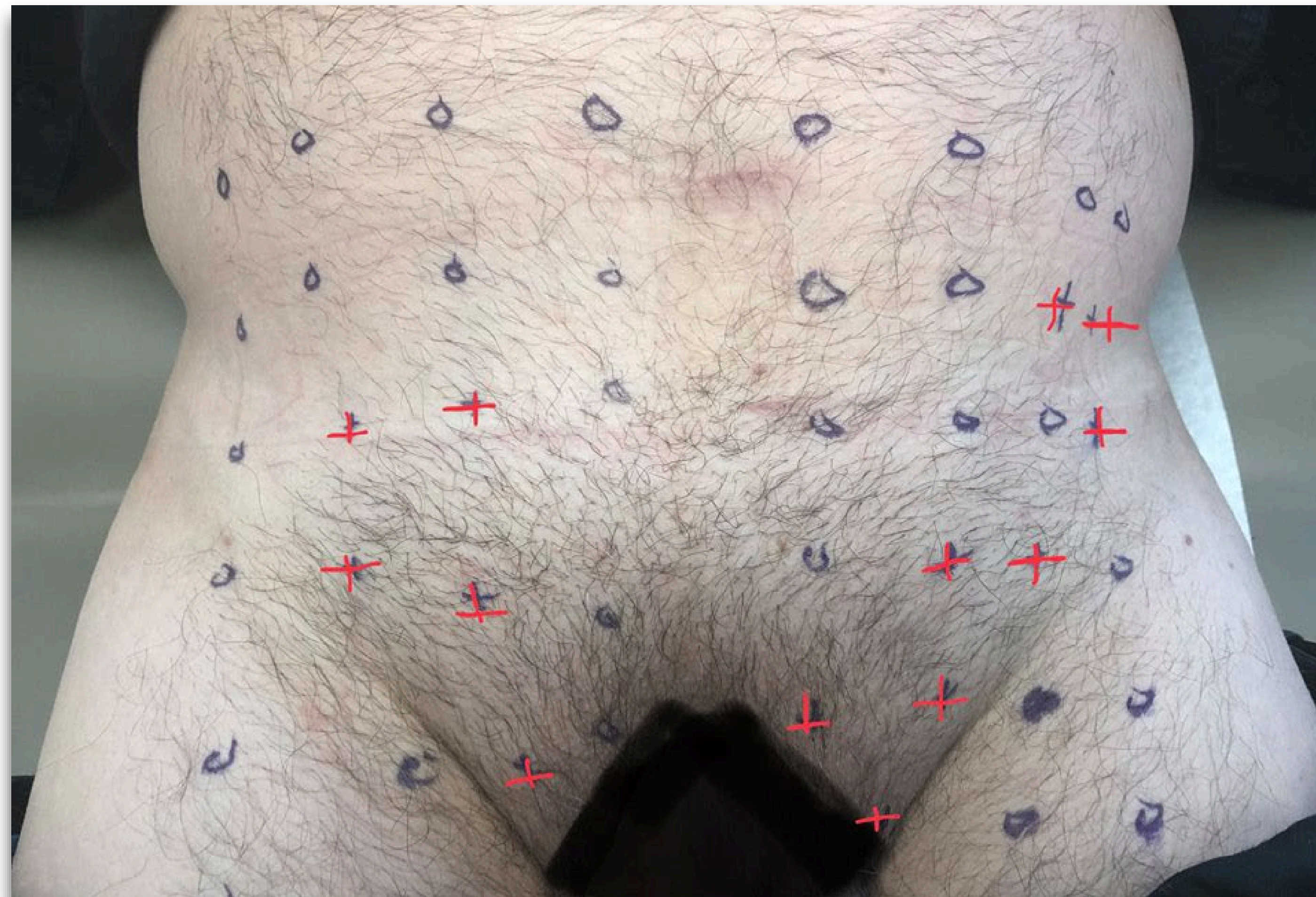
Secondary Groin Pain

- Nociceptive pain: caused by activation of nociceptors by nociceptive molecules due to tissue injury or inflammatory reaction. These signals are then transmitted to the brain via A-delta and C-fibers.
- Neuropathic pain: caused by direct nerve injury due to direct contact of nerves with mesh and/or nerve entrapment by sutures, staples, tacks, folded mesh, or meshoma.

Chronic Groin pain



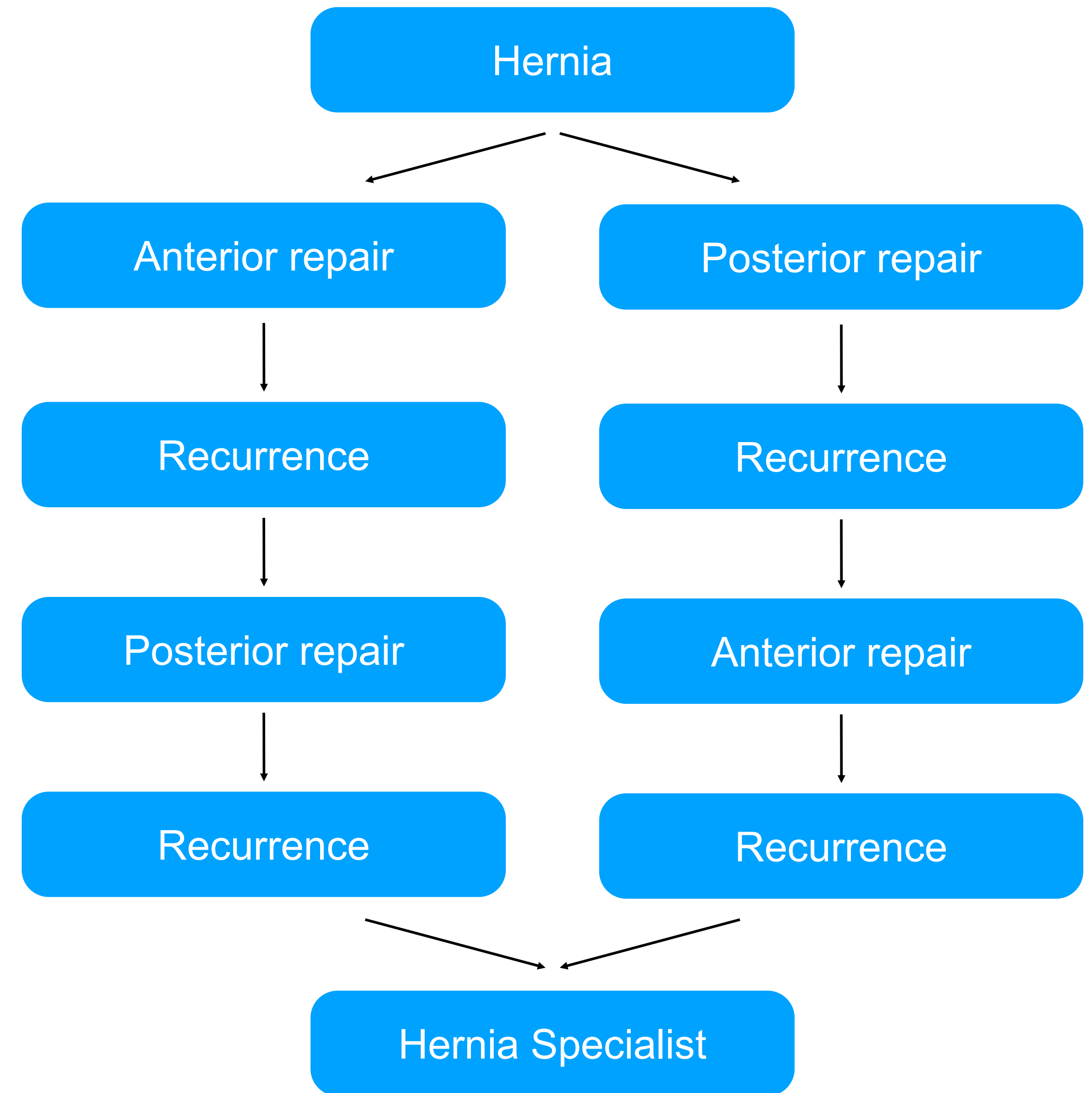
<http://www.groinpainclinic.co.uk/groin-pain/nerves-affected-in-chronic-groin-pain>

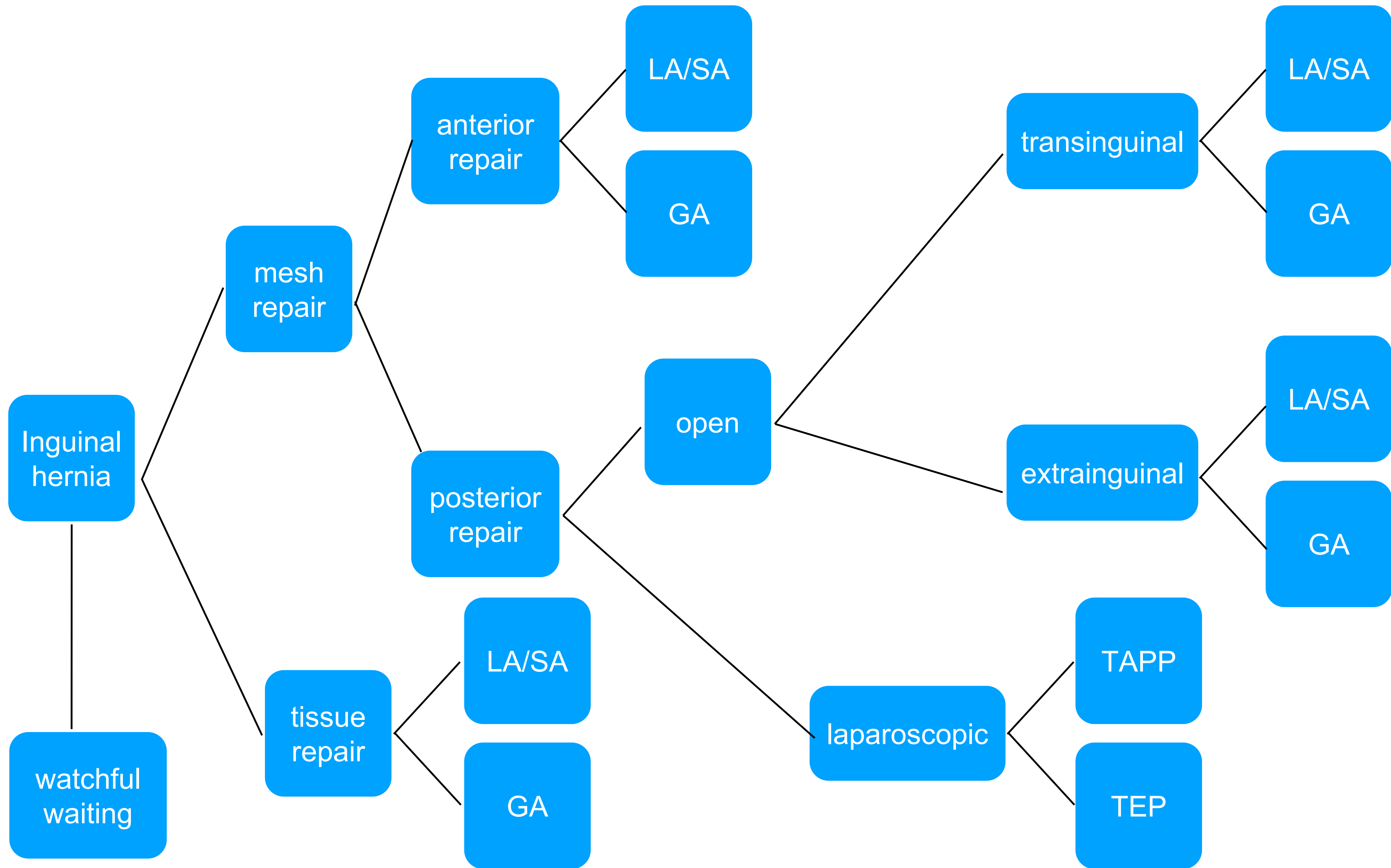


<https://www.facebook.com/photo.php?fbid=10102235878598208&set=gm.1371261833013793&type=3&eid=ARBu01Nuv5XEWGxq1tigSdp-RdN28tJoT6rst0ZKNtMyFZjUF8ZdSoRqP0Ap-W-85FlerqLLbwvvhBuY&ifg=1>

Laparoscopic repair for all recurrences?

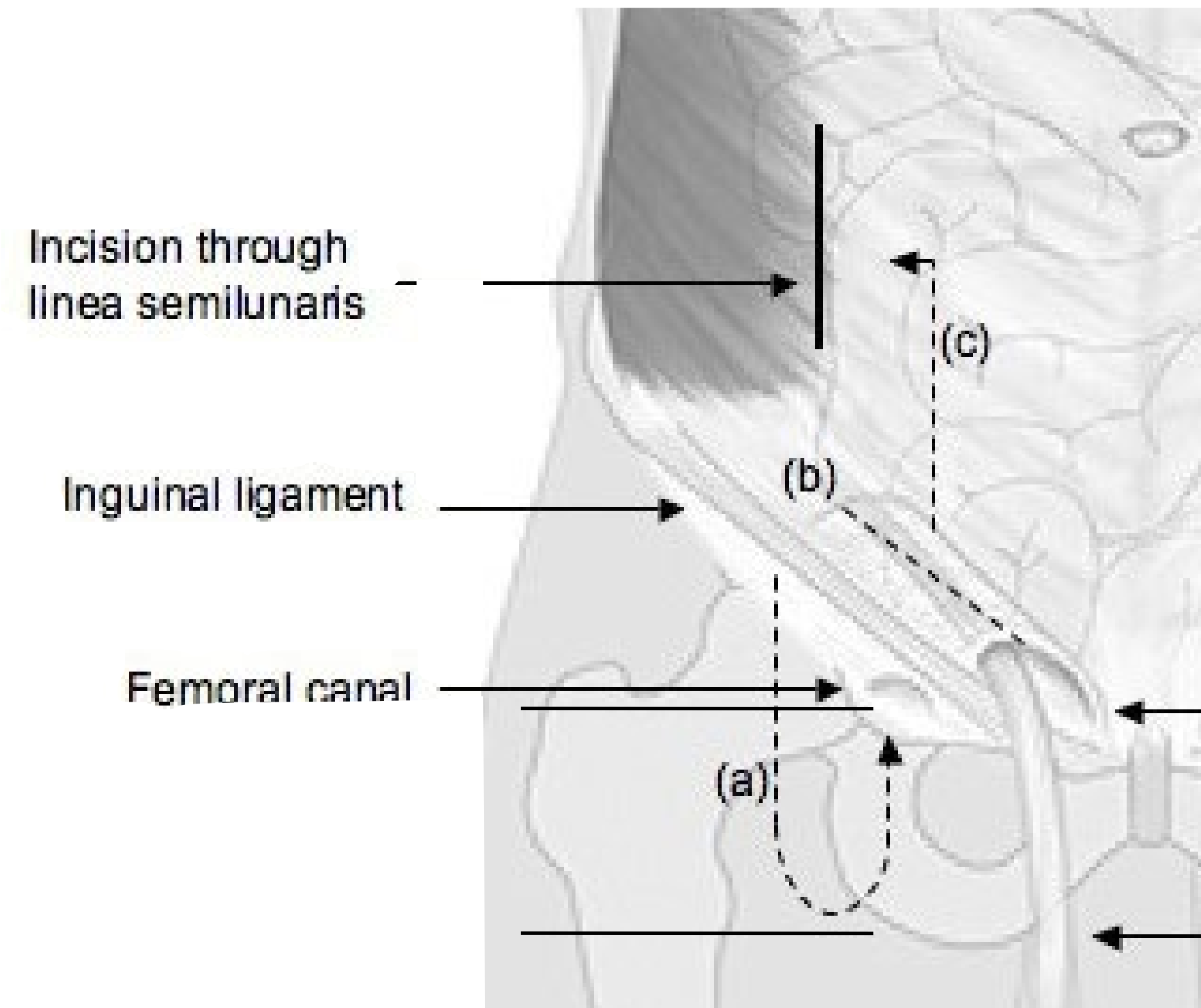
avoiding scar tissue in recurrent hernias by choosing a different approach





Femoral Hernia

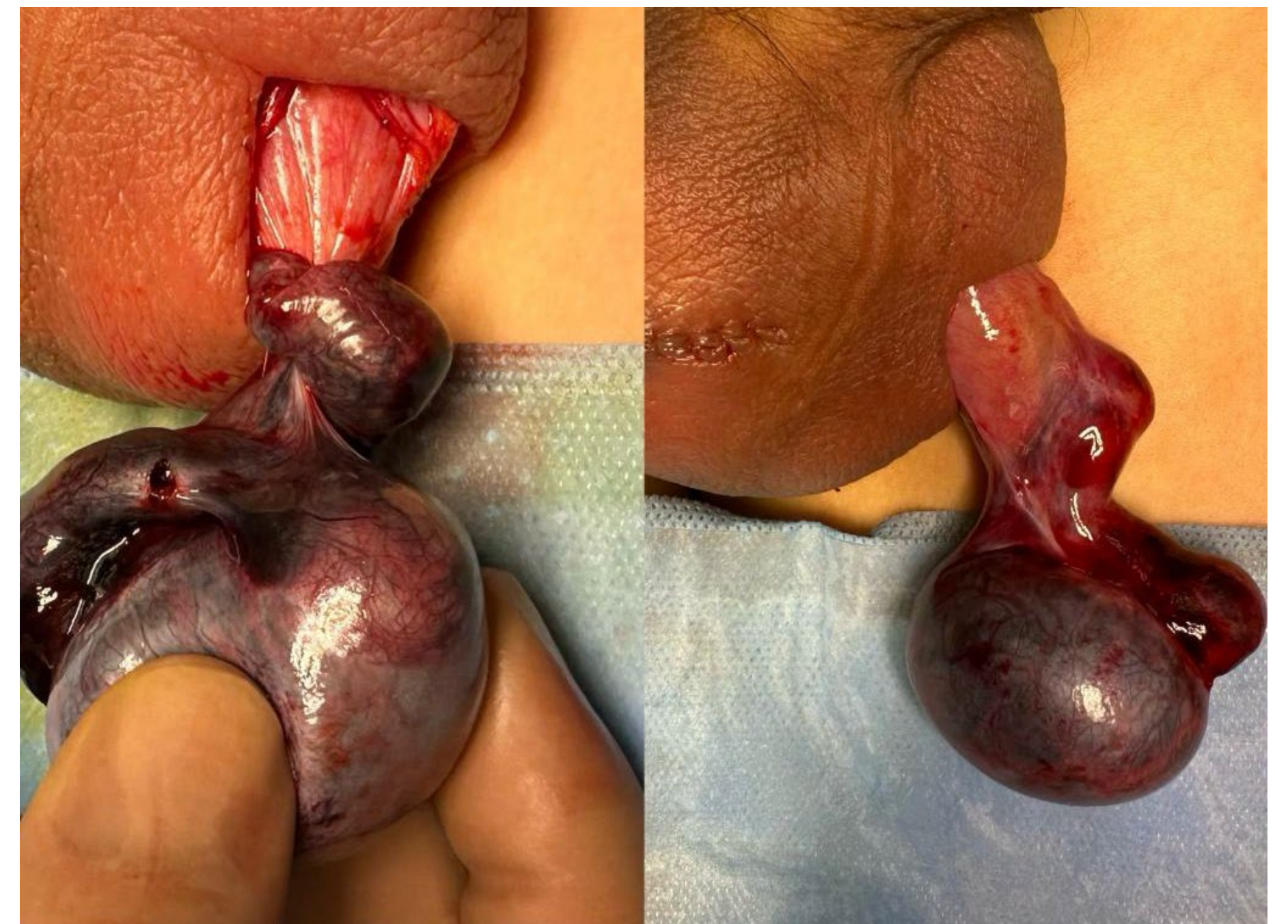
- (a) Lockwood's infrainguinal approach
- (b) Lotheissen's transinguinal approach
- (c) McEvedy's high approach



Depends whether there is strangulation

Testicular Torsion

- Pediatric emergency
- Unilateral scrotal pain often associated with vomiting
- Surgical Emergency



Hydrocele

- Fluid in the tunic vaginalis
- Can get above it
- Transillumination
- Fluctuence+
- Testis not palpable
- USG to examine the testis
- Primary vs. secondary hydrocele



Primary Hydrocele

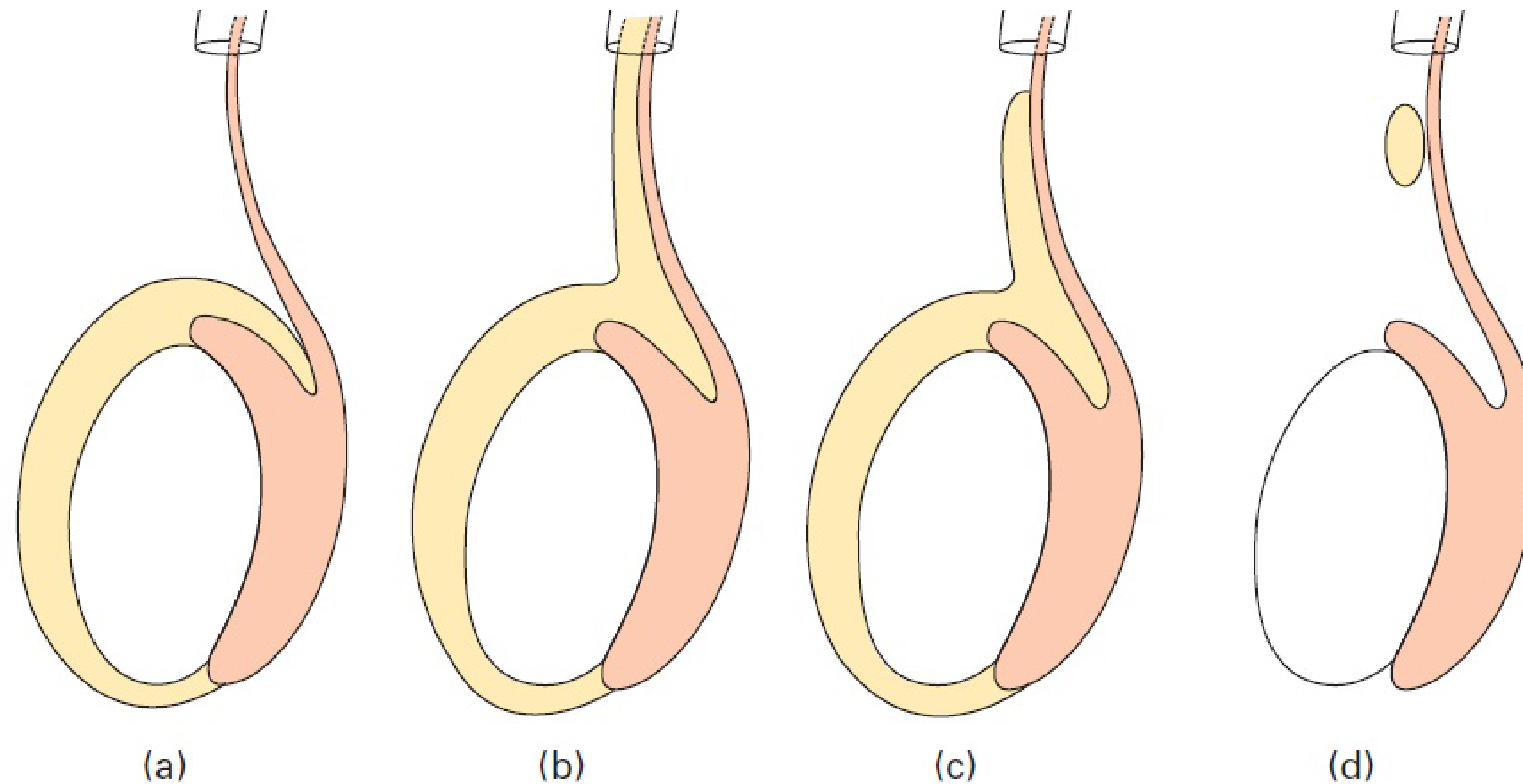
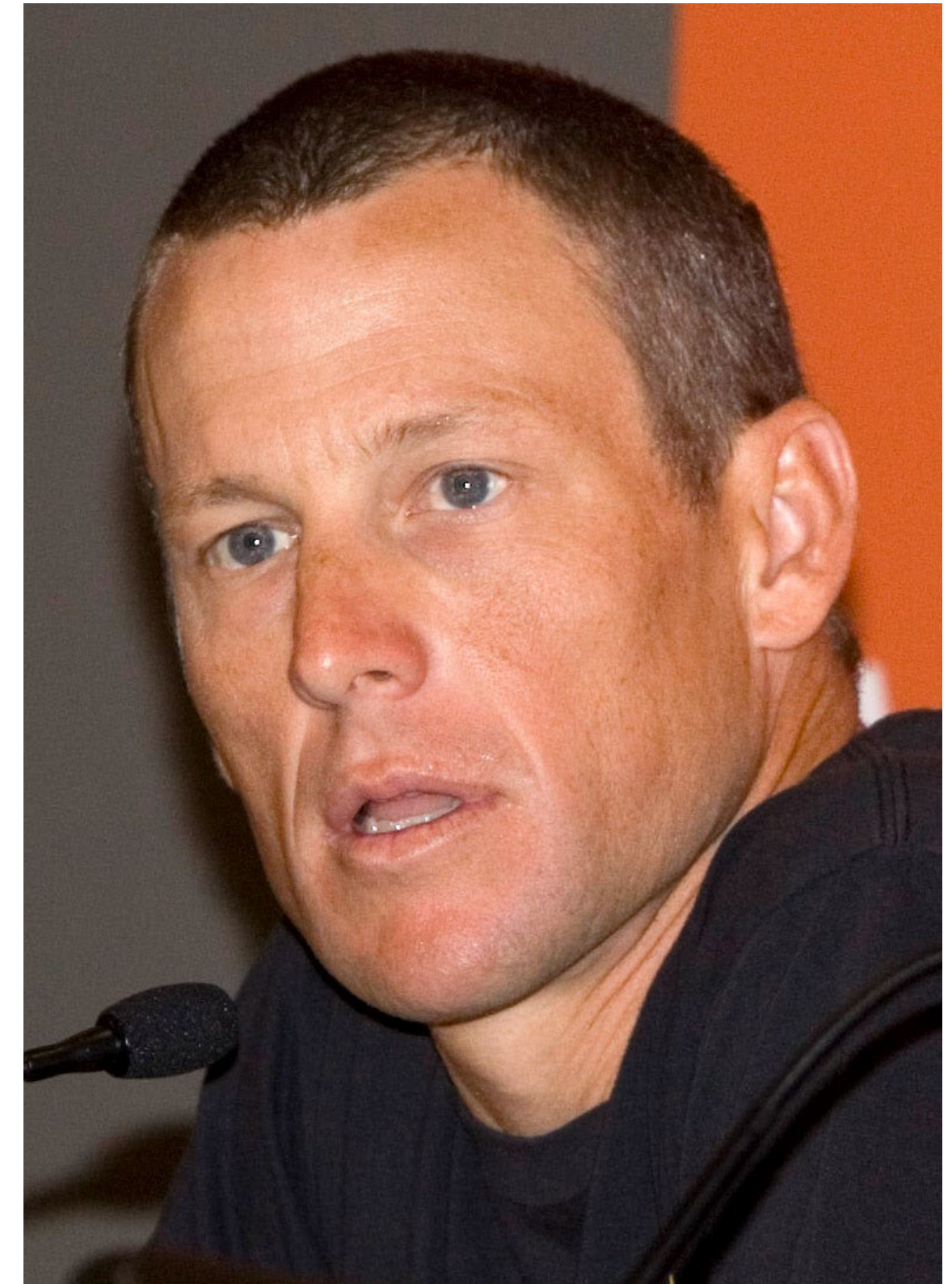


Fig. 91 Types of hydrocele. (a) Vaginal hydrocele, (b) congenital hydrocele, (c) infantile hydrocele, (d) hydrocele of the cord. (The tube at the upper end of each diagram represents the internal inguinal ring. Yellow = hydrocele, Brown = vas and epididymis)

<http://www.biswaforum.com/hydrocele-of-testis/>

Management

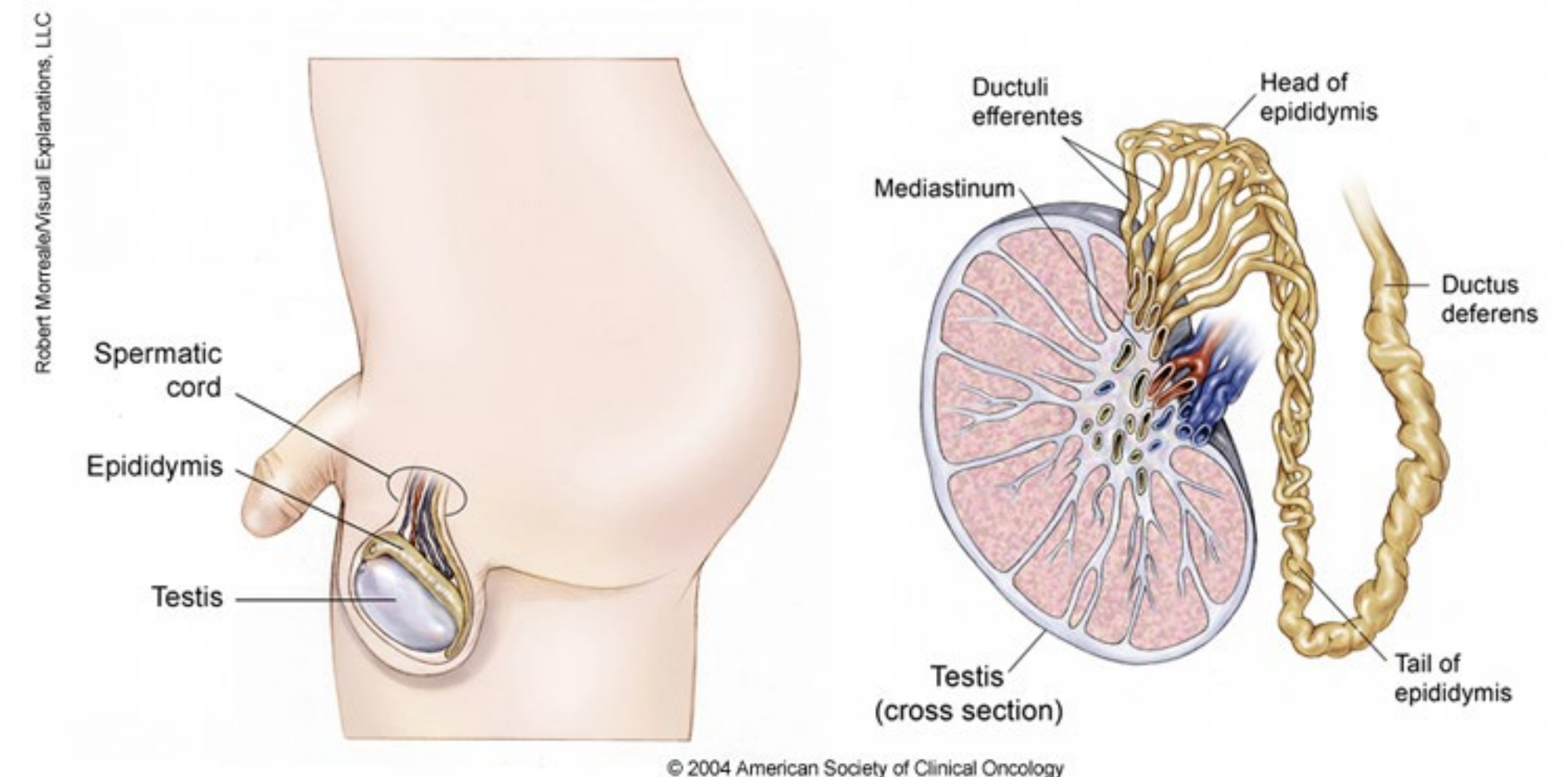
- Needle aspiration
- Jaboulay procedure



25/M % testicular swelling, hemoptysis

Testicular tumor

- Risk Factor: 20-45yo, undescended testis, family & personal history, white, HIV
- Enlarged, firm & non-tender testis
- AFP, hCG, LDH-1
- Classified according to the cell type
- Inguinal radical orchidectomy ± RPLND



Systemic Chemotherapy

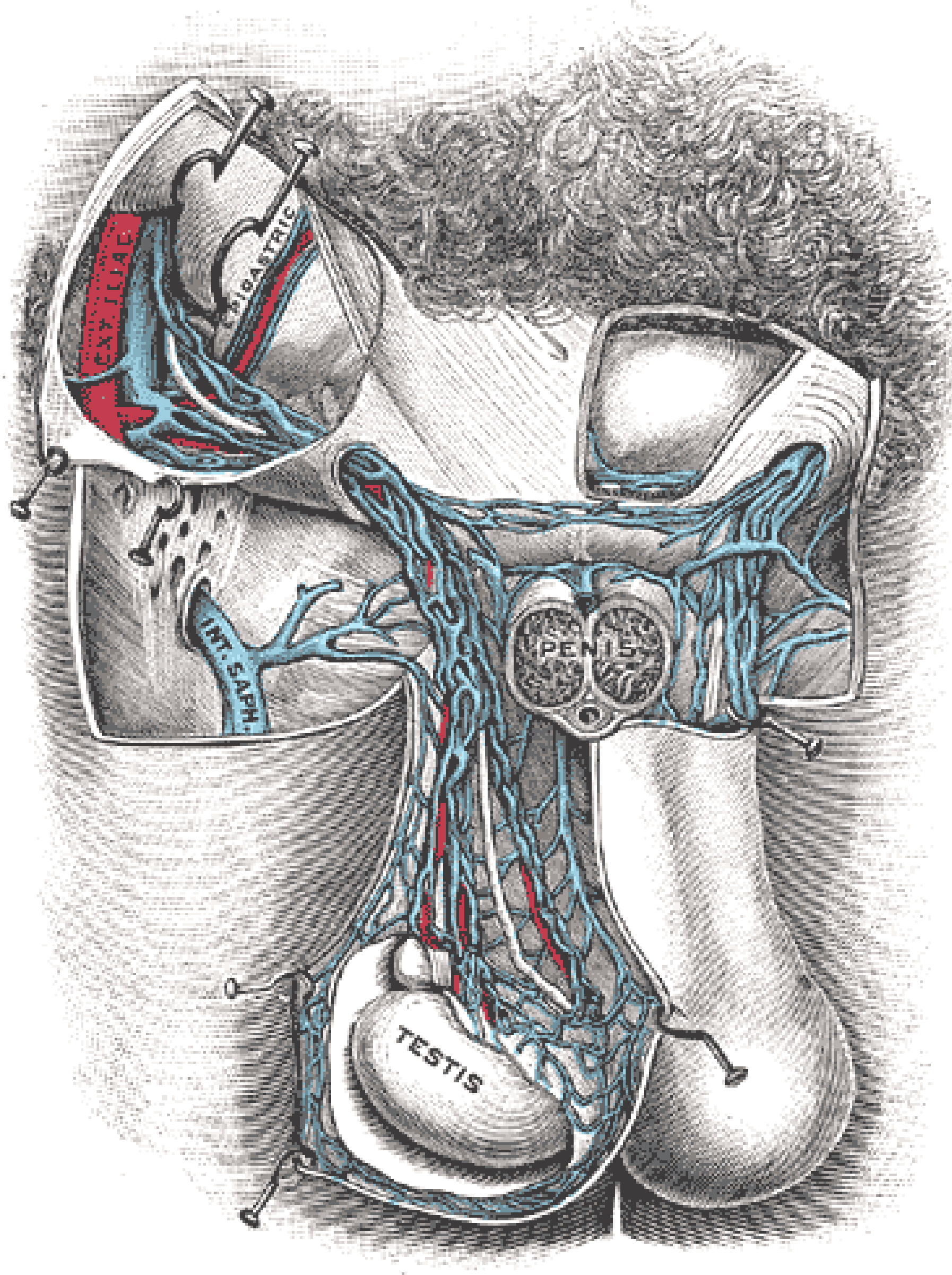
Regimens

- BEP: bleomycin, etoposide, and cisplatin
- Carboplatin (for stage I pure seminoma only)
- EP: etoposide and cisplatin
- TIP: paclitaxel, ifosfamide, and cisplatin
- VeIP: vinblastine, ifosfamide, and cisplatin
- VIP: etoposide, ifosfamide, and cisplatin
- High-dose carboplatin and etoposide
- Bleomycin (Blenoxane)
- Carboplatin (Paraplatin)
- Cisplatin (Platinol)
- Etoposide (Toposar, VePesid)
- Gemcitabine (Gemzar)
- Ifosfamide (Cyfos, Ifex, Ifosfamidum)
- Oxaliplatin (Eloxatin)
- Paclitaxel (Taxol)
- Vinblastine (Velban, Velsar)

WHO Classification

- **Germ cell tumor**
 - Seminoma Spermatocytic seminoma
 - Embryonal carcinoma and teratoma (teratocarcinoma)
 - Teratoma (Mature, Immature, With malignancy transformation)
 - Choriocarcinoma
 - Yolk sac tumor
- **Sex cord stromal tumors**
 - Leydig cell tumor
 - Sertoli cell tumor
 - Granulosa cell tumor
 - Mixed forms

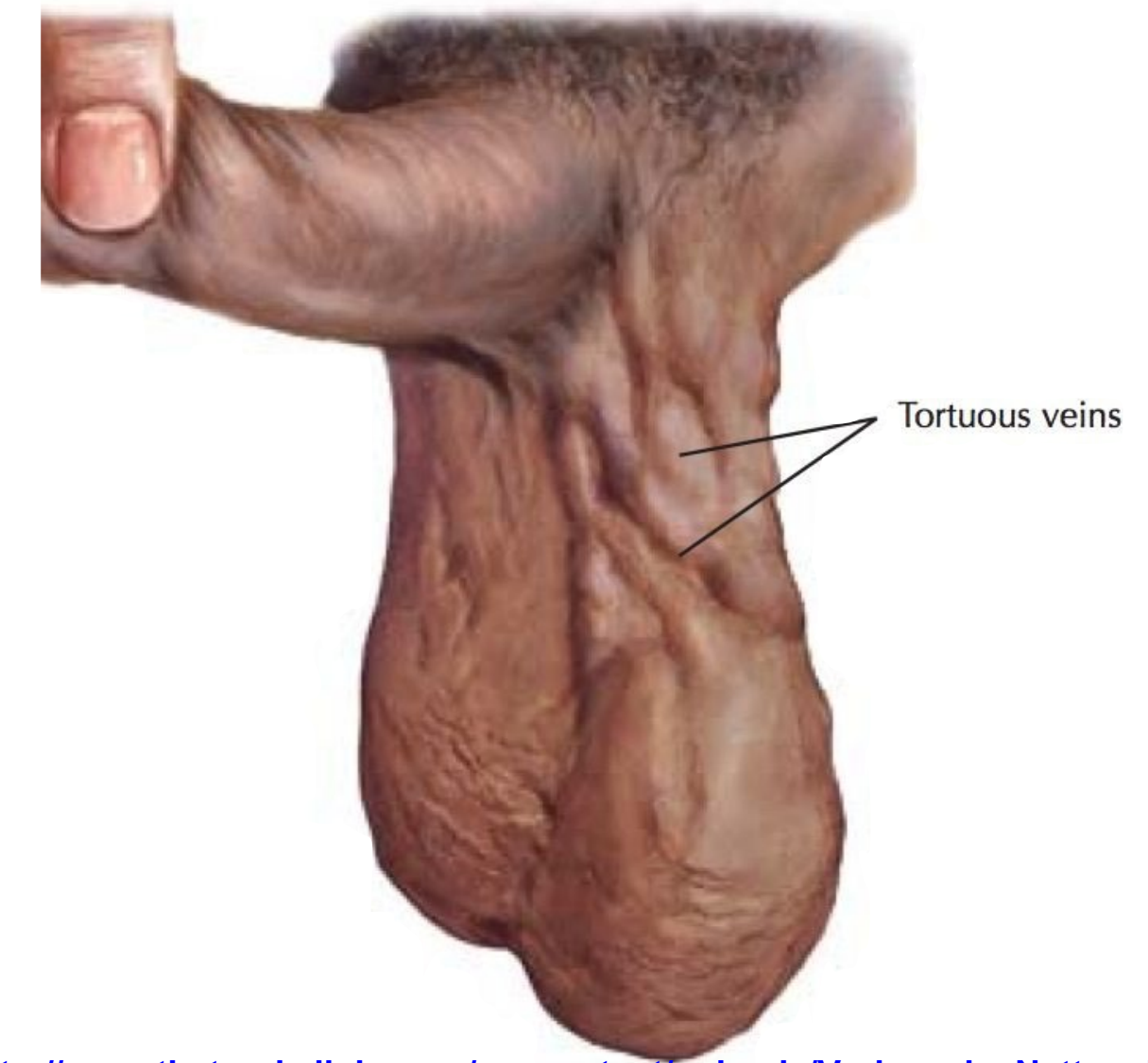
Varicocele



- Pampiniform venous plexus of drains blood from the testicles
- Abnormal enlargement of the pampiniform venous plexus in the scrotum.
- Occurs in around 15% of men.
- Left side more common than right.
- May lead to issues with fertility

Varicocele

- Bag of worms felt on standing
- Rule out renal mass causing venous congestion
- Nutcracker syndrome
- Usually presents with scrotal swelling, soft lumps.
- Occasional present with pain, and heaviness in their scrotum.



<http://www.theturekclinic.com/wp-content/uploads/Varicocele.-Netters.jpg>



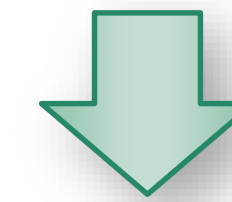
By James Heilman, MD - Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=14660935>

The evaluation questionnaire(s) for the MBBS Whole Class Session(s)

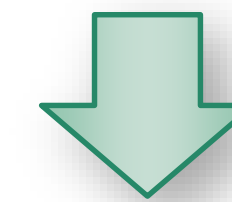
<https://www.bimhse.hku.hk/intranet/Evaluation>

Thank you! Questions?

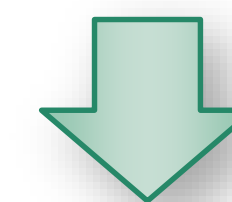
yipjeremy@hku.hk



Login with your Portal ID



Click the questionnaire for WCS



Find my name (Jeremy Yip)